Metaorders, Price Impact, and Crowding: Evidence from Proprietary and Client Flow

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Abstract

Large trading decisions are typically split into many smaller trades and executed through market members, either for the member's own account or on behalf of clients. This paper asks whether these two trading capacities generate the same market footprint. Using trade-by-trade data for FTSE MIB constituents from June 2024 to May 2025, with aggressive trades attributed to members and labelled as proprietary or client flow, we reconstruct member-level metaorders and compare their execution characteristics, price impact, impact dynamics, and crowding environment. Proprietary and client metaorders are similar in their broad statistical properties: both display heterogeneous, heavy-tailed distributions of size, duration, and relative volume. The main execution-level difference is that proprietary metaorders tend to trade with higher participation rates. The price response, however, differs more sharply. Client metaorders produce an impact curve close to the square-root law and their impact partly reverts after execution. Proprietary metaorders display a more concave impact law, higher fitted impact over most of the observed size range, and a more persistent post-execution footprint. Crowding reveals the mechanism only in part. Client flow is more strongly aligned with same-day client imbalance, especially at high participation rates, so daily crowding is not by itself the source of the proprietary impact premium. The relevant distinction appears at the execution-time scale: proprietary metaorders are more often surrounded by overlapping flow in the same direction. Controlling for this active directional overlap weakens the proprietary premium. Overall, the evidence shows that proprietary and client metaorders should not be pooled: trading capacity matters for impact, persistence, and the interpretation of crowded order flow.

1 Introduction

Understanding how large orders move prices is central both to execution-cost analysis and to theories of price formation. In modern electronic markets, however, large trading decisions typically reach the market through intermediating members, who may execute on behalf of clients or trade for their own account. This distinction is economically meaningful: proprietary and client trades may differ in information content, execution style, inventory motives, and exposure to crowding, with potentially different consequences for both the level and the persistence of market impact. Yet most empirical studies of metaorders do not cleanly separate these two trading capacities. Using trade-by-trade data for FTSE MIB constituents in which aggressive trades are attributed to individual members and labelled as proprietary or client, this paper exploits that separation to compare the properties, price impact, and crowding of proprietary and client metaorders within a common empirical framework.

A robust empirical regularity in market microstructure is the non-linear, concave relation between the average price impact of large orders and their size, measured either in volume or in execution time. When a metaorder of total size Q shares is executed over a day by slicing it into many smaller trades, the average price change Δp typically scales as a power law,

$$\Delta p = Y\epsilon\sigma_D\left(\frac{Q}{V_D}\right)^{\gamma}, \qquad 0 < \gamma < 1, \tag{1}$$

where $\epsilon = \pm 1$ is the sign of the trade, σ_D is the daily volatility and V_D the daily volume. The constant Y captures the execution quality and might depend, for example, on the trading algorithm. The "square-root law" arises when $\gamma \approx 1/2$ and it has been observed across markets and asset classes . This behavior has been documented in a wide range of institutional datasets and synthetic metaorder reconstructions, and is remarkably robust to variations in the microstructure, i.e. tick size, asset type, etc [6, 1, 2, 3, 4, 5, 23].

At the same time, institutional trading activity is highly crowded [20, 15, 16]: different investors often trade in the same direction on the same names and days, which amplifies execution costs through co-impact. In co-impact models, the aggregate price response is driven by the joint distribution of metaorder signs and sizes, with correlations in order flow leading to a non-linear impact surface and to important differences between idiosyncratic and crowded trades [21, 19].

The present analysis focuses on a high-frequency dataset where trade-by-trade aggressive activity can be attributed to individual members and separated into proprietary and client flows. Using these data, we:

1. Reconstruct metaorders as same-sign sequences of aggressive trades at the member level;
2. Document the empirical distributions of their durations, sizes, relative sizes in units of daily volume, and execution participation rates;
3. Estimate a volatility-normalised power-law impact function as a function of the metaorder's size;
4. Study crowding measures for proprietary and client flows.

Against this background, we investigate the following questions:

1. Do proprietary and client metaorders differ systematically in their basic execution characteristics, such as size, duration, participation rate, and distribution across members?
2. Do proprietary and client metaorders obey the same impact law, or do they differ in the scale and concavity of price impact once size and impact are properly normalised?
3. Do their dynamic impact profiles differ, in particular in the build-up of impact during execution and in the decay or persistence after completion?
4. How does crowding differ across proprietary and client flow, both within each group and in their cross-group interaction? Does crowding help explain the observed differences in impact between proprietary and client metaorders?

The remainder of the paper is organized as follows. Section 2 describes the trade-level data and the metaorder reconstruction procedure. Section 3 documents the descriptive properties of proprietary and client metaorders, including their distributional features, execution schedules, and member-nationality composition. Section 4 estimates and compares their price-impact laws and dynamic impact paths. Section 5 studies daily, cross-group, active-overlap, and member-level crowding, and evaluates whether these mechanisms explain the proprietary–client impact differences. Section 6 concludes, while Appendix A reports additional results for Italian executing members.

2 Data and Metaorder Reconstruction

This section describes the underlying trade-level dataset and the procedure used to reconstruct member-level metaorders. We also define the per-metaorder quantities used throughout the descriptive analysis, impact estimation, and crowding results.

2.1 Trade-level data

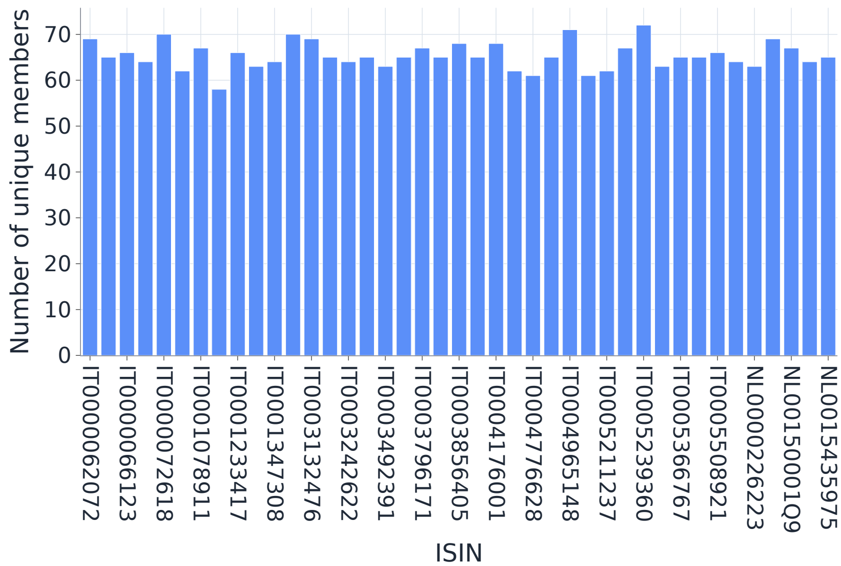
We investigate an event-level sequence of aggressive trades for all the FTSE MIB constituents between June 6, 2024 and May 30, 2025 for a total of 251 trading days The number of investigated stocks is 41.

For each trade j we have:

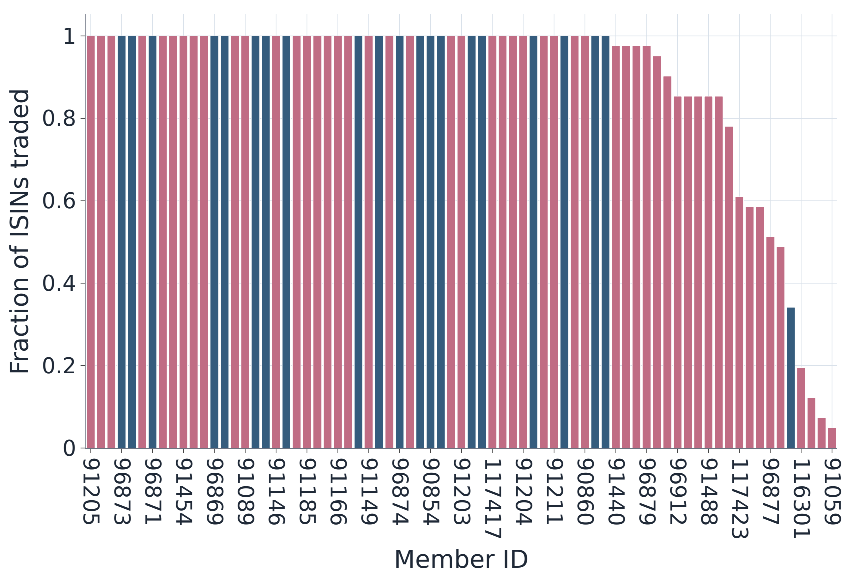
- an instrument identifier (ISIN) $k(j)$;
- a timestamp t_j;
- an aggressive direction $\varepsilon_j \in \{+1, -1\}$, with $\varepsilon_j = +1$ for aggressive buys and $\varepsilon_j = -1$ for aggressive sells;
- a traded volume $q_j > 0$ (number of shares);
- the first and last execution prices P_j^f, P_j^l;
- an aggressive capacity flag distinguishing proprietary ("dealing on own account") from client (non-proprietary) trading;
- a client identifier and a member identifier.

Only trades in the continuous trading session are used (from 09:30 to 17:30) and trades are sorted by time within each ISIN to form a time-ordered event stream. As the goal of this project is to study the differences and the interplay between proprietary and client metaorders, it is useful to collect some descriptive statistics about these two kinds of trades and the overall behavior of the market members. There are 73 active members across all ISINs in the sample, and Figure 2a shows the number of active members per ISIN. The figures indicate that almost all members are active in almost all ISINs and that the majority of them traded all the assets, with a very small fraction focused on a smaller subset of the ISINs.

Across ISINs, the ratio between proprietary and client aggressive trades is relatively stable, as shown in Figure 2a. Proprietary aggressive trades are consistently more numerous than client trades, with a ratio ranging from 0.60 to 1.30.

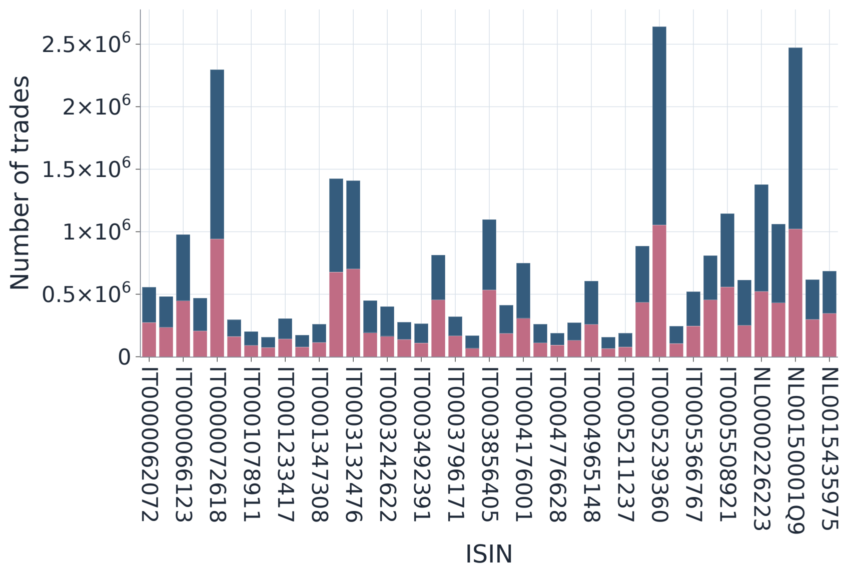


(a) Active members per ISIN.

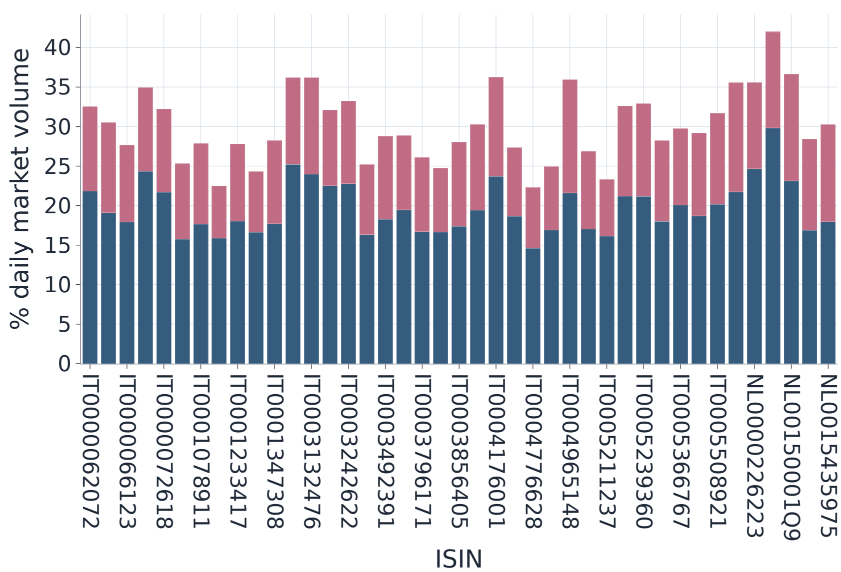


(b) ISIN coverage per member (IT: blue, foreign: red).

Figure 1



(a) Number of proprietary and client trades per ISIN (blue: proprietary, red: client).



(b) Mean percentage of volume within a metaorder with respect to the overall daily volume (blue: proprietary, red: client.)

Figure 2

2.2 Metaorder definition at the member level

Before detailing how we construct metaorders from our dataset, it is useful to briefly review the main approaches that have been proposed in the literature to identify such metaorders. Our data set is particularly valuable in this respect, as it provides access to both client and member identifiers, as well as detailed information on the nature of each trade. These features substantially improve the accuracy of our metaorder reconstruction, placing it just below datasets in which metaorders are explicitly labeled. In summary, the literature distinguishes four broad categories of metaorder detection methods:

- Labelled metaorders: use datasets in which parent orders or execution packages are explicitly reported, so metaorders can be observed directly after applying sample filters [14, 18, 22];
- Detection of same-sign sequences from proxies of trader IDs: infer persistent trading agents from available identifier proxies, then group their consecutive aggressive trades with the same sign into candidate metaorders [23];
- Detection of same-sign sequences from unlabeled member data: reconstruct hidden order splitting by aggregating sequences of same-direction trades associated with the same intermediary, without observing the underlying client identity [11, 12];
- Detection of same-sign sequences from proprietary data with labels on the nature of trades: exploit the distinction between proprietary and client flow to construct and compare metaorders separately across trading categories [13];
- Identification from regimes in the order flow: detect persistent directional regimes or changes in latent order-flow imbalance, and interpret these regimes as periods of coordinated buying or selling pressure [8];
- Synthetic metaorder construction: generate artificial metaorders calibrated to observable market activity when real parent-order identifiers are unavailable, allowing controlled impact analysis under known construction rules [24];

Metaorder construction We work at the member level: a metaorder is a sequence of trades routed by a given member, for a given client in a given ISIN and day, with constant sign of aggressive activity. The construction proceeds as follows.

Fix an instrument k and a member a. Let $I_{a,k}$ be the ordered set of indices j such that member a trades instrument k at time $t_j \in [09{:}30, 17{:}30]$. For each $j \in I_{a,k}$ we observe the sign ε_j and volume q_j.

We first detect runs of constant sign:

$$M = \{j_s, j_s + 1, \ldots, j_e\} \subset I_{a,k} \tag{2}$$

such that

$$\varepsilon_j = \varepsilon_{j_s}, \qquad \forall j \in M, \tag{3}$$

and M is maximal with this property (no strictly larger contiguous set has a constant sign). We apply several filters to the detected sequences:

- A sequence is kept if $|M| \geq 5$

- All trades in the run must belong to the same calendar day; runs spanning the market close are discarded.

- All trades in the run share the same client identifier; this ensures that we do not aggregate orders from different final investors.

Even after this step, very long runs can contain long periods of inactivity. We finally split runs on large gaps. Let $\Delta t_k = t_{j_{k+1}} - t_{j_k}$ be successive inter-trade gaps within a run. If $\Delta t_k > 1$ hour, the run is split at that point into separate metaorders. Again, metaorders with fewer than five trades are discarded. This yields a collection of same-day, same-client, same-sign metaorders for each ISIN.

Each resulting metaorder i is characterised by:

- its set of child trades $M_i = \{j_s(i), \ldots, j_e(i)\}$,

- its sign $\varepsilon_i \in \{+1, -1\}$ (direction), taken as the sign of its trades,

- its start and end timestamps $t_i^{\text{start}} = t_{j_s(i)}$ and $t_i^{\text{end}} = t_{j_e(i)}$ defining the execution interval $[t_i^{\text{start}}, t_i^{\text{end}}]$,

- the capacity flag (proprietary vs client (non-proprietary)).

We thus obtain two large metaorder datasets:

- proprietary metaorders (group "prop"),

- client (non-proprietary) metaorders (group "client") [1]

Table 1 summarises the number of metaorders for each group before and after every filter.

Category	**Initial Number**	**After Min Trades**	**After Min Duration**
Proprietary	2,774,525	866,437	588,543
Client	988,240	308,026	256,371

Table 1: Metaorder counts for proprietary vs client flow across filtering steps (gap split at 1 hour, min trades set to 5, min duration to 120 seconds).

The final dataset contains 588, 543 proprietary metaorders and 256, 371 client metaorders, thus an average of 57 and 25 proprietary and client metaorders per day and ISIN, respectively. Buy and sell metaorders are of comparable frequency in both groups (proprietary: 279,992 buys vs. 308,551 sells; client: 129,919 buys vs. 126,452 sells) and the share of daily volume associated with a metaorder ranges between 0.22 and 0.42 with a systematically higher contribution from proprietary metaorders, see Figure 2b.

[1] It is crucial to emphasize that a portion of trades labeled as client trades are in fact institutional executions carried out by entities with direct market access on behalf of end-clients who lack such access. These are not retail trades.

2.3 Metaorder features

For each metaorder i we compute:

- Metaorder volume
$$Q_i = \sum_{j \in M_i} q_j, \tag{4}$$
i.e. the total number of shares executed by the metaorder.

- Metaorder direction.
$$\varepsilon_i = \begin{cases} +1 & \text{buy metaorder} \\ -1 & \text{sell metaorder} \end{cases} \tag{5}$$

- Duration
$$T_i = t_i^{\text{end}} - t_i^{\text{start}}. \tag{6}$$

- Inter-arrival times. Within each metaorder, gaps between consecutive child trades are
$$\Delta t_i^{(\ell)} = t_{j_{\ell+1}} - t_{j_\ell}, \qquad \ell = 1, \dots, |M_i| - 1. \tag{7}$$

- Average Daily volume. In order to smooth day–to–day fluctuations, we consider the 5-day average daily volume
$$\bar{V}_d = \frac{1}{K_d} \sum_{k=0}^{K_d - 1} V_{d-k}, \tag{8}$$
where $K_d = 5$ and V_d is the total volume traded in that ISIN on day d

- Average Daily volatility. For each day, we compute the intraday integrated volatility σ_d using a realised-kernel estimator [7]. Then, we average it similarly as the average daily volume
$$\bar{\sigma}_d = \frac{1}{K_d} \sum_{k=0}^{K_d - 1} \sigma_{d-k}, \tag{9}$$
The sampling frequency used in the estimation was selected based on the signature plots, choosing the point beyond which the volatility estimates become stable. Using this approach, we decided to sample the data every 1000 seconds. This method differs from the one commonly used in the literature (for example, in [23]), where the volatility is taken as the difference between the opening and closing prices of the day.

- Relative size (fraction of daily volume).
$$\phi_i = \frac{Q_i}{V_{d(i)}} \in (0, 1], \tag{10}$$
where $d(i)$ denotes the day of metaorder i. This is the fraction of the daily traded volume executed by the metaorder and enters in the square-root impact law.

- Participation rate within the execution window. Let V_i^{window} denote the total market volume traded in the same instrument between t_i^{start} and t_i^{end}. The execution participation rate is
$$\eta_i = \frac{Q_i}{V_i^{\text{window}}}. \tag{11}$$
This is the metaorder's market share during its own execution.

- Start and end prices. P_i^{start} and P_i^{end}, taken as volume-weighted prices at the corresponding timestamps [2].

3 Descriptive Properties of Metaorders

We now present the basic stylised facts about the reconstructed metaorders, focusing on the first research question presented in the introduction: Do proprietary and client metaorders differ systematically in their basic execution characteristics, such as size, duration, participation rate, and distribution across members?.

Unless stated otherwise, the results are computed at the member level and aggregated across ISINs and days.

3.1 Distributions

In this section, we examine the distributions of several key metrics, most of which will be used in the following sections to fit metaorder price impact. For each metric, we consider the decay behaviour of the tail of its distribution. Specifically, we apply the power-law fitting procedure introduced in [9], which is widely regarded as the standard approach for fitting power-law models to empirical data and benchmarking them against plausible alternative distributions. In our analysis, we compare the power-law fit with lognormal, truncated power-law, and exponential alternatives.

Before turning to the main analysis, it is useful to examine how metaorders are distributed across members and whether some members specialize in executing metaorders rather than isolated, sparse trades. Figure 3 addresses both questions. The rank plot of N_m across members is broad, revealing a small number of highly active members that execute a very large number of metaorders, alongside a long tail of members with much lower metaorder activity. The scatter plot, however, does not suggest any clear specialization in metaorder execution. Rather, it indicates that some members execute relatively fewer metaorders compared with their overall trading activity than others.

The metrics analyzed, together with the corresponding results shown in Figure 4, are:

- metaorder duration T,
- inter-arrival times Δt between child orders (expressed in minutes),
- metaorder volume Q,
- relative metaorder size ϕ,
- participation rate η.

We find no striking differences in the statistical properties of the distributions between client and proprietary metaorders. The most notable distinction appears in the distribution of participation rates: proprietary metaorders tend to shift the mass of the distributoin towardas larger values of η. Overall, all metrics span several orders of magnitude and display heavy-tailed distributions, consistent with the heterogeneous execution styles observed in other metaorder datasets [11, 12, 14].

As we can see from Table 2 and Figure 4, no distributions showed an actual power law tail behaviour. For all the metrics, the best fit is either a lognormal or a truncated power law model, with a significant tie between the two only for the durations T and relative sizes ϕ in the case of proprietary metaorders. Table 3 reports the fitted parameters of the first two best models together

[2] A single trade can hit multiple price levels, even if it is rare. In those cases, we consider the volume-weighted price

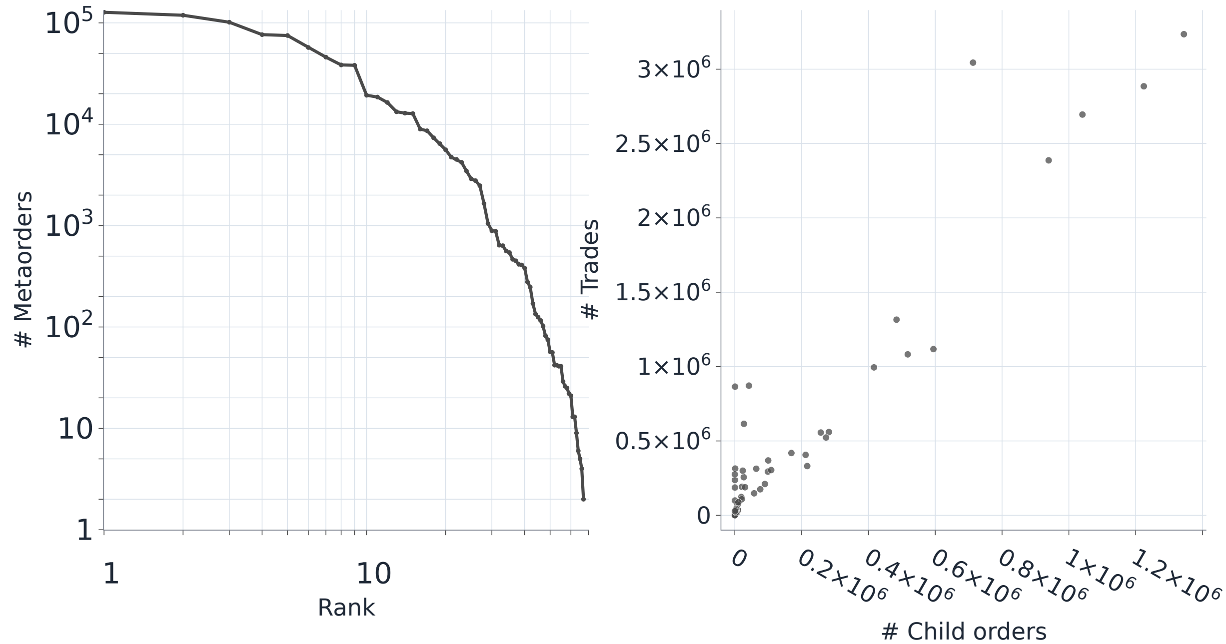


Figure 3: Rank plot of the number of metaorders per member (left) and scatter plot of the total number of child orders across all metaorders versus the total number of metaorders at the member level (right).

with the corresponding percentile bootstrap confidence intervals. However, to compare our results with the literature, we overlay on each distribution also the power law fit with the corresponding decaying exponent.

	Best Fit by AIC		2nd Best		$\log \frac{\mathcal{L}_{1^{\text{st}}}}{\mathcal{L}_{2^{\text{nd}}}}$		p-value	
	Client	Proprietary	Client	Proprietary	Client	Proprietary	Client	Proprietary
T	truncated power law	lognormal	lognormal	truncated power law	47.48	4.78	~ 0	0.5511
Δt	truncated power law	truncated power law	lognormal	lognormal	2930.58	2431.79	~ 0	~ 0
Q	truncated power law	truncated power law	lognormal	lognormal	47.53	305.10	~ 0	~ 0
Q/V	truncated power law	lognormal	lognormal	truncated power law	5.05	80.07	0.2511	~ 0
η	lognormal	truncated power law	truncated power law	exponential	26.24	67.08	0.0082	~ 0

Table 2: Best vs second fit review by metric and group. The p values corresponds to the significance test performed on the differences in log likehood between the best and second best models. Values below 10^{-5} are shown as ~ 0.

3.2 Execution schedule characterization

The distributional evidence above already points to an important difference between the two groups: proprietary metaorders display systematically larger participation rates than client metaorders. Since participation is tightly linked to execution aggressiveness, this raises a natural question: are the two groups also characterized by different execution schedules, beyond differences in size and duration alone? This question is also motivated by the dynamic impact results of Section 4.3 that

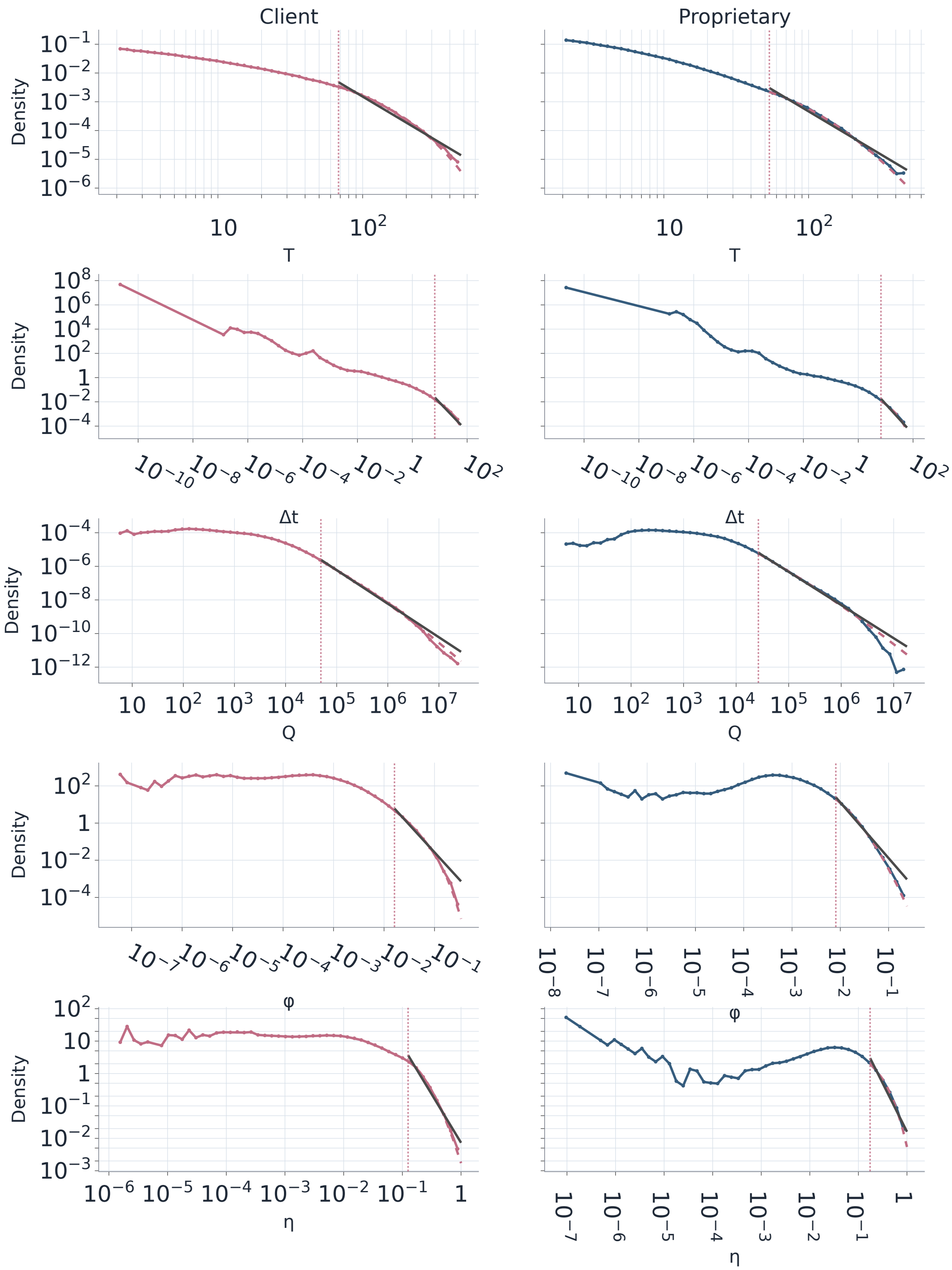


Figure 4: Empirical probability density functions of selected metaorder metrics, conditioned on trading capacity. Vertical dashed lines indicate the estimated lower bound of the power-law regime in the right tail. Solid black lines represent the best power-law fits. The estimated scaling exponents, alongside the statistically preferred model among the tested alternatives, are reported in the legend.

Metric	Client model	Client parameters (95% CI)	Proprietary model	Proprietary parameters (95% CI)
T	truncated power law	$x_{\min} = 68.0812$ $[62.2530, 68.1113]$ $\alpha = 0.9926$ $[0.8344, 1.0501]$ $\lambda = 0.0120$ $[0.0115, 0.0130]$	lognormal	$x_{\min} = 53.5471$ $[48.5295, 52.1330]$ $\mu = 3.5921$ $[3.5287, 3.6328]$ $\sigma = 0.7805$ $[0.7643, 0.7995]$
Δt	truncated power law	$x_{\min} = 6.5999$ $[5.2875, 10.0567]$ $\alpha = 0.9774$ $[0.5661, 1.0875]$ $\lambda = 0.0580$ $[0.0541, 0.0714]$	truncated power law	$x_{\min} = 6.7493$ $[6.2979, 8.4972]$ $\alpha = 1.2348$ $[1.0791, 1.2727]$ $\lambda = 0.0551$ $[0.0537, 0.0605]$
Q	truncated power law	$x_{\min} = 49139.0000$ $[30919.3750, 57036.0000]$ $\alpha = 1.8760$ $[1.8548, 1.8905]$ $\lambda = 2.0 \times 10^{-7}$ $[1.8 \times 10^{-7}, 2.4 \times 10^{-7}]$	truncated power law	$x_{\min} = 26532.0000$ $[25076.0750, 32605.9000]$ $\alpha = 1.8598$ $[1.8504, 1.8713]$ $\lambda = 3.2 \times 10^{-7}$ $[3.0 \times 10^{-7}, 3.5 \times 10^{-7}]$
ϕ	truncated power law	$x_{\min} = 0.0161$ $[0.0072, 0.0167]$ $\alpha = 1.9200$ $[1.7238, 2.0055]$ $\lambda = 23.9470$ $[21.4535, 28.7676]$	lognormal	$x_{\min} = 0.0081$ $[0.0069, 0.0082]$ $\mu = -5.7773$ $[-5.8757, -5.7264]$ $\sigma = 0.9545$ $[0.9367, 0.9801]$
η	lognormal	$x_{\min} = 0.1251$ $[0.1147, 0.1258]$ $\mu = -2.1999$ $[-2.2194, -2.1726]$ $\sigma = 0.6815$ $[0.6699, 0.6900]$	truncated power law	$x_{\min} = 0.1740$ $[0.1683, 0.1693]$ $\alpha = 0.3102$ $[0.2961, 0.3661]$ $\lambda = 6.4970$ $[6.3697, 6.5553]$

Table 3: Preferred-model parameter estimates by metric and group, together with the estimated lower cutoff $x_{\min}$. Confidence intervals are percentile bootstrap intervals at the 95% level from the nonparametric full-sample bootstrap with 50 replicates; each replicate re-estimates the lower cutoff and refits all candidate models.

	Sample Size		Mean		Median	
	Client	Proprietary	Client	Proprietary	Client	Proprietary
T	256371	588543	36.8402	19.1341	19.0411	8.8617
Δt	3449403	5297419	2.7381	2.1258	0.5665	0.4085
Q	256371	588543	38933.1611	30800.1527	7172.0000	6519.0000
Q/V	256371	588543	0.0042	0.0034	0.0018	0.0018
η	256371	588543	0.0844	0.1362	0.0499	0.0949

Table 4: Distribution means and medians by metric and group.

we anticipate here: if proprietary impact is more persistent, one possible channel is that proprietary metaorders are executed in a different temporal pattern.

To address this point, we store the child-trade schedule of every metaorder. Let $t_{i,\ell}$ and $q_{i,\ell}$ denote, respectively, the timestamp and size of child trade ℓ of metaorder i. Using the previously defined start and end times t_i^{start} and t_i^{end}, we define the normalized execution time of each child trade as

$$u_{i,\ell} = \frac{t_{i,\ell} - t_i^{\text{start}}}{t_i^{\text{end}} - t_i^{\text{start}}} \in [0,1], \tag{12}$$

and its normalized volume weight as

$$w_{i,\ell} = \frac{q_{i,\ell}}{Q_i}. \tag{13}$$

The cumulative execution schedule of metaorder i is then

$$C_i(u) = \sum_{\ell: u_{i,\ell} \leq u} w_{i,\ell}, \qquad u \in [0,1]. \tag{14}$$

By construction, $C_i(0) = 0$ and $C_i(1) = 1$. For descriptive comparison, we summarize each group $G \in \{\text{prop}, \text{client}\}$ by the pointwise median cumulative schedule

$$\widetilde{C}_G(u) = \text{med}_{i \in G}\, C_i(u). \tag{15}$$

For reference, a TWAP execution corresponds to the diagonal $C(u) = u$.

Figure 5 reports the median cumulative schedule together with the heatmap of the empirical density of $C_i(u)$ in a single object: within each panel, the solid line traces the group median and the dashed diagonal gives the TWAP benchmark. The clearest result is that both groups are front-loaded relative to TWAP. By $u = 0.10$, the median cumulative fraction is 0.224 for proprietary metaorders and 0.206 for client metaorders. Equivalently, proprietary metaorders reach one quarter of their final volume at $u \simeq 0.136$, whereas client metaorders reach the same threshold at $u \simeq 0.151$. This points to somewhat earlier execution for proprietary flow at the start of the metaorder, but the gap is modest.

The cross-group difference remains small over the rest of the schedule. Around the middle of the execution, the two median curves are very close, with $\widetilde{C}_{\text{prop}}(0.50) = 0.549$ and $\widetilde{C}_{\text{client}}(0.50) = 0.559$. In the second half of the metaorder, client flow slightly overtakes proprietary flow: by $u = 0.75$ the median cumulative fractions are 0.758 and 0.776, and by $u = 0.90$ they are 0.893 and 0.907, respectively. We therefore read the evidence as showing broadly similar schedule shapes, with only a mild tendency for proprietary metaorders to execute slightly earlier and for client metaorders to catch up later. Since this subsection is descriptive and does not attach formal inference to the median-gap estimates, we do not interpret these differences as evidence of a strong statistical separation between the two groups.

The strength of this analysis is that it uses the full child-trade path of the metaorder sample, rather than a reduced-form scalar statistic, and does so on a very large panel of 588,334 proprietary and 255,535 client metaorders. At the same time, its main limitation is that it is deliberately descriptive. The median curve still pools together heterogeneous names, directions, participation rates, and execution motives; for this reason it should not be interpreted as a structural estimate of an "optimal" schedule. The heatmap in Figure 5 is useful precisely because it makes this dispersion visible: behind the median path, execution styles are far from homogeneous. We therefore read the schedule evidence as a characterization of the temporal footprint of proprietary and client flow, not as a causal statement about the origin of the difference.

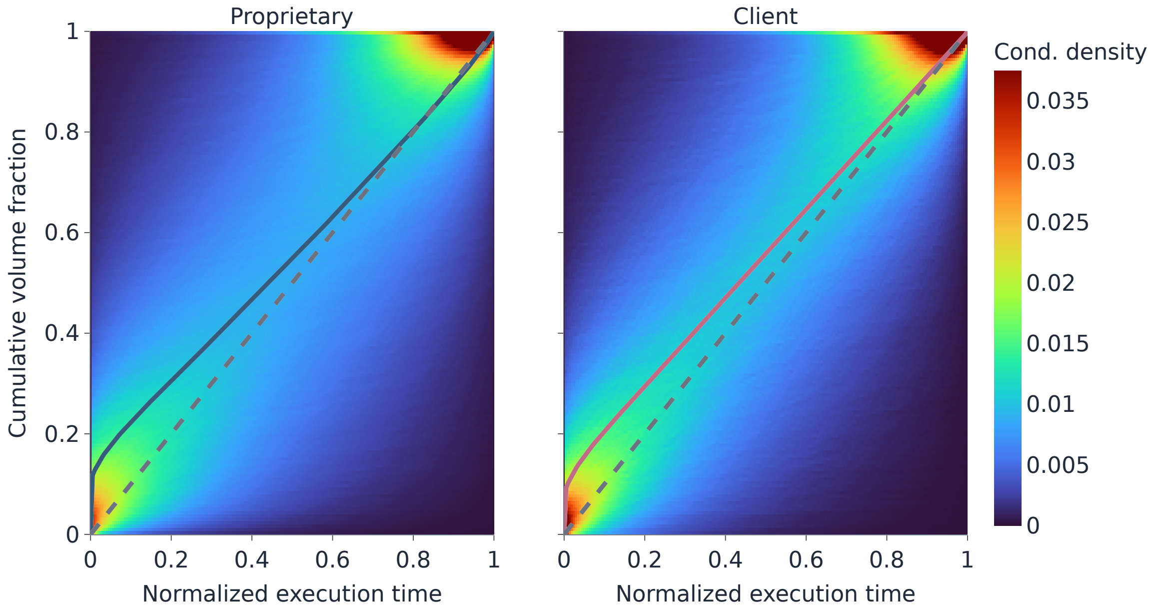


Figure 5: Execution schedule characterization at the member level. Left: proprietary metaorders. Right: client metaorders. Each panel reports the conditional density of cumulative schedules as a function of normalized execution time; the solid line overlays the group median cumulative schedule and the dashed diagonal is the TWAP benchmark. The heatmap makes the cross-sectional heterogeneity around the median path directly visible.

3.3 Nationality composition of executing members

At the member level, we can assign each metaorder to an Italian (IT) or foreign executing broker using the member nationality mapping. The comparison with overall trading activity helps clarify the result. In proprietary flow, Italian members are already marginal in the market as a whole, accounting for only 1.94% of traded volume, and they are even less represented in the metaorder sample, with 0.36% of metaorders (2,093) and 0.57% of metaorder volume. Only 11.00% of Italian proprietary volume is executed through detected metaorders, compared with 38.09% for foreign members. In client flow, the message is different: Italian members are not small overall, since they account for 52.72% of traded volume, but they contribute only 9.40% of client metaorders (24,111) and 9.77% of client metaorder volume. Equivalently, only 4.15% of Italian client volume is executed through detected metaorders, against 42.69% for foreign members. Thus, the dominance of foreign brokers in the metaorder sample is not merely a reflection of larger aggregate activity; it also reflects a much stronger tendency to execute through metaorder-like order splitting, especially on the client side. Across both flows combined, Italian members account for 26.34% of traded volume but only 3.10% of detected metaorders. Aggregate distributions, impact estimates, and crowding results reported in this paper are therefore driven predominantly by foreign-member execution.[3]

Given this extreme skew, nationality-conditioned results must be interpreted cautiously, especially for proprietary IT flow. We nevertheless report in Section 4.3 the dynamic-impact paths for the Italian subsample, while the remaining Italian-member summary statistics are collected in Appendix A.

[3]Nationality refers to the executing member (broker), not necessarily to the final investor.

Flow	IT	Foreign
Proprietary	2,093 (0.36%)	586,450 (99.64%)
Client (non-proprietary)	24,111 (9.40%)	232,260 (90.60%)

Table 5: Member nationality split at the member level (FTSEMIB sample). Percentages are computed on metaorders with known member nationality.

4 Impact Model

To compare metaorders across instruments and dates, we normalise sizes by daily volume and impacts by a daily volatility proxy. Building on these normalised quantities, we estimate an average impact law as a function of relative size and study how it varies between proprietary and client flow.

4.1 Instantaneous impact in volatility units

The instantaneous impact of metaorder i in volatility units is defined as

$$I_i = \frac{\varepsilon_i \, \Delta p_i}{\sigma_{d(i)}}. \tag{16}$$

By construction, I_i is typically positive on average (buys push the price up, sells push it down), but individual realisations can be negative because of noise and concurrent market activity.

Before fitting, we apply a minimum-size filter $\phi_i > 10^{-5}$ to avoid the extremely noisy region of small metaorders. We also exclude the few metaorders with participation rate equal to one by imposing $\eta_i < 1$. These additional two filters remove, respectively, $\sim 0.12\%$ and $\sim 0.02\%$ from the entire metaorder sample.

This yields a cleaned dataset of (ϕ_i, I_i) pairs with auxiliary features (participation, capacity, etc.) for subsequent analysis.

4.2 Impact models

Following [14], we posit two different functional forms for the relation between the expected instantaneous impact and the relative size:

$$\begin{aligned} &\mathbb{E}[I_i \mid \phi_i] = Y\,\phi_i^{\gamma} && \text{Power law} \\ &\mathbb{E}[I_i \mid \phi_i] = a\log_{10}(1 + b\phi_i) && \text{Logarithm} \end{aligned} \tag{17}$$

where:

- $Y > 0$ is a scale parameter (impact for unit participation);
- $\gamma \in (0, 1)$ is the impact exponent (with $\gamma = 1/2$ corresponding to a square-root law [10, 23]);
- $a, b > 0$ are the parameters of the logarithmic model, which serves as a benchmark to assess deviations from power-law behaviour (e.g. saturation or linear regimes).

Taking logarithms of the power-law specification yields:

$$\log \mathbb{E}[I_i \mid \phi_i] = \log Y + \gamma \log \phi_i. \tag{18}$$

Because individual I_i are noisy and often negative, we do not regress directly on single metaorders. Instead we proceed via log-binning.

Binning in relative size. We form K logarithmic bins in $\log_{10}\phi$ over the range of the data. For each bin k we compute:

- the bin centre ϕ_k (e.g. geometric mean of ϕ_i in the bin);
- the mean impact $\bar{I}_k$ over metaorders in the bin;
- the standard error of the mean SEM_k.

Bins with fewer than a prescribed minimum count, non-positive or non-finite $\bar{I}_k$, or non-finite SEM_k are discarded.

Estimation procedure: Power Law. For the power-law model, we define

$$X_k = \log\phi_k, \qquad Z_k = \log\bar{I}_k. \tag{19}$$

We fit the linear model

$$Z_k = \alpha + \gamma X_k + \varepsilon_k, \tag{20}$$

where $\alpha = \log Y$ and γ is the impact exponent. The fit is performed by weighted least squares (WLS) in log-space, using weights inversely proportional to the variance of the bin estimate [10]:

$$w_k \approx \left(\frac{\bar{I}_k}{\mathrm{SEM}_k}\right)^2. \tag{21}$$

Bins with more metaorders and smaller dispersion are given larger weight, while very noisy or poorly populated bins contribute less. This yields estimates $\hat{\alpha}$ and $\hat{\gamma}$ together with their standard errors; the scale parameter is then recovered as $Y = e^{\hat{\alpha}}$.

Estimation procedure: Logarithmic Benchmark. Alongside the power law, we fit the logarithmic specification $\bar{I}_k = a\log_{10}(1 + b\phi_k)$ on the same binned statistics $(\phi_k, \bar{I}_k)$. Unlike the power law, this model is estimated via a non-linear fit on the bin means. Comparing the two curves allows us to visually inspect deviations from the power law, particularly at the extremes of the size distribution where the logarithmic form imposes a linear behaviour for $\phi \to 0$ and a slower saturation for very large ϕ.

Empirical impact curves. Applying this procedure separately to proprietary and client metaorders yields the impact curves in Figure 6 and Figure 6, where the two groups are overlaid in the same panel to facilitate comparison. The WLS estimates for the power law specification are:

- proprietary: $Y = 0.07264 \pm 0.00366$, $\gamma = 0.3662 \pm 0.0083$;
- client: $Y = 0.1281 \pm 0.0154$, $\gamma = 0.5082 \pm 0.0228$.

Two differences stand out. First, the scale parameter for client flow is substantially larger than for proprietary flow: $Y_{\text{client}}/Y_{\text{prop}} \approx 1.76$. This parameter, however, should not be read in isolation as a ranking of fitted impact, because the exponent also differs sharply across groups. Since $\gamma_{\text{prop}} < \gamma_{\text{client}}$, the proprietary curve is flatter in log-log scale and can lie above the client curve for sufficiently small relative sizes. In the present fit, the two curves cross at about $\phi^\star \approx 1.9 \times 10^{-2}$: below this threshold, which covers most of the empirical range, proprietary metaorders have the larger fitted

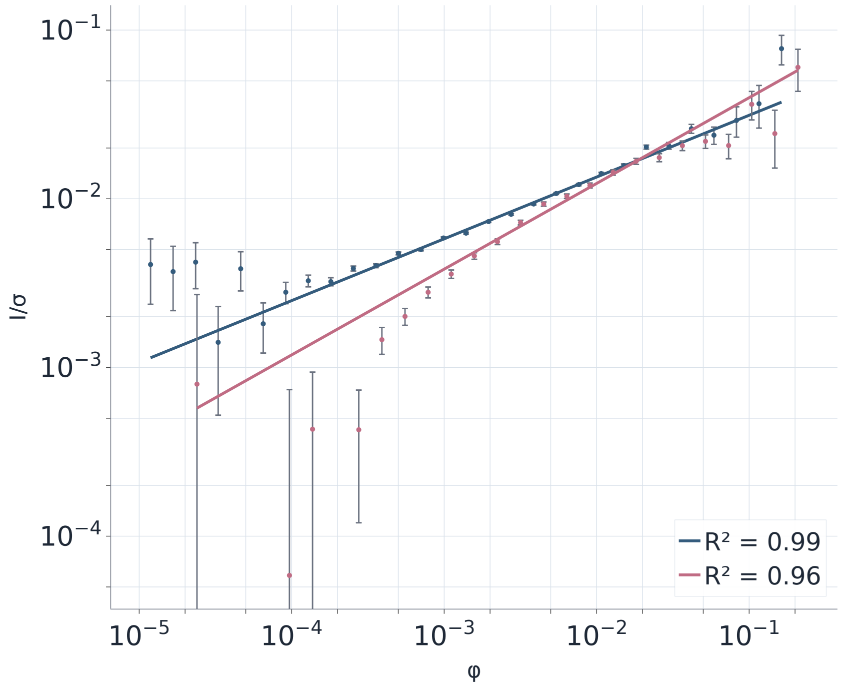


Figure 6: Overlay of proprietary and client overall impact fits for the power law specification (markers: bin means ± SEM; blue: proprietary, red: client).

impact; above it, the client curve overtakes. Second, the client exponent is close to the square-root benchmark, whereas proprietary flow displays a markedly more concave scaling.

For client flow, the logarithmic specification provides an excellent fit ($R^2 = 0.99$) while the power law remains a competitive description ($R^2 = 0.96$). For proprietary flow, by contrast, the power law provides the best fit ($R^2 = 0.99$) than the logarithmic model ($R^2 = 0.92$). Splitting by side yields very similar parameters for proprietary buys and sells, so for brevity we focus on the pooled estimates.

The analysis can be refined by conditioning on participation rate η_i. Splitting metaorders into quantiles of η_i and repeating the log-binned WLS within each quantile yields a family of impact curves

$$\mathbb{E}[I \mid \phi, \eta \in \text{quantile } q] = Y_q \phi^{\gamma_q},$$

summarized in Figure 8. For proprietary flow, the conditional fits remain close, with a mildly larger exponent at low participation ($\gamma = 0.38$) than at high participation ($\gamma = 0.35$). For client flow, the dependence on participation is much stronger: the low-η subset displays a much steeper scaling ($\gamma = 0.84$) than the high-η subset ($\gamma = 0.39$), so the two conditional curves can cross over the ϕ range. This behaviour highlights a non-separable interaction between relative size and participation, consistent with the joint $(Q/V, \eta)$ dependence reported in [14].

4.3 Market impact trajectory during and after the execution

The scalar impact measure I_i in the previous section captures only the price change between the first and last trade of a metaorder. To obtain a time–resolved view of how impact builds up during execution and relaxes afterwards, we also track the full impact path of each metaorder.

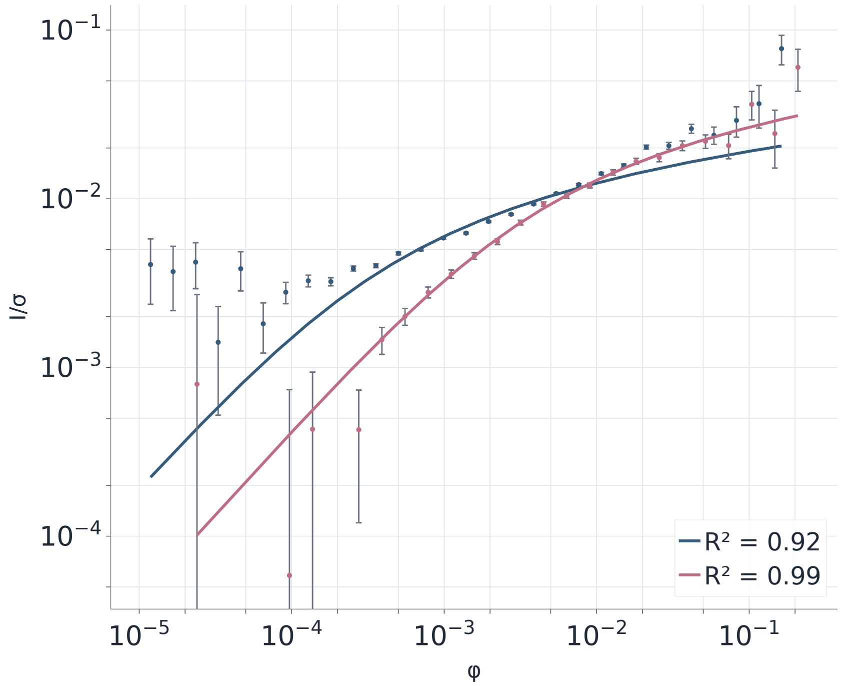


Figure 7: Overlay of proprietary and client overall impact fits for the logarithmic specification (markers: bin means ± SEM; blue: proprietary, red: client).

For a given metaorder i on day $d(i)$, using the previously defined start and end times t_i^{start} and t_i^{end} and duration T_i, we define the instantaneous normalised impact at calendar time $t \geq t_i^{\text{start}}$ as

$$I_i(t) = \frac{\varepsilon_i}{\sigma_{d(i)}} \left(\log P(t) - \log P_i^{\text{start}} \right), \tag{22}$$

This is the same normalisation used for the end-of-execution impact, but evaluated along the whole trajectory of the price process. We then introduce a normalised time variable τ at the metaorder level. During execution,

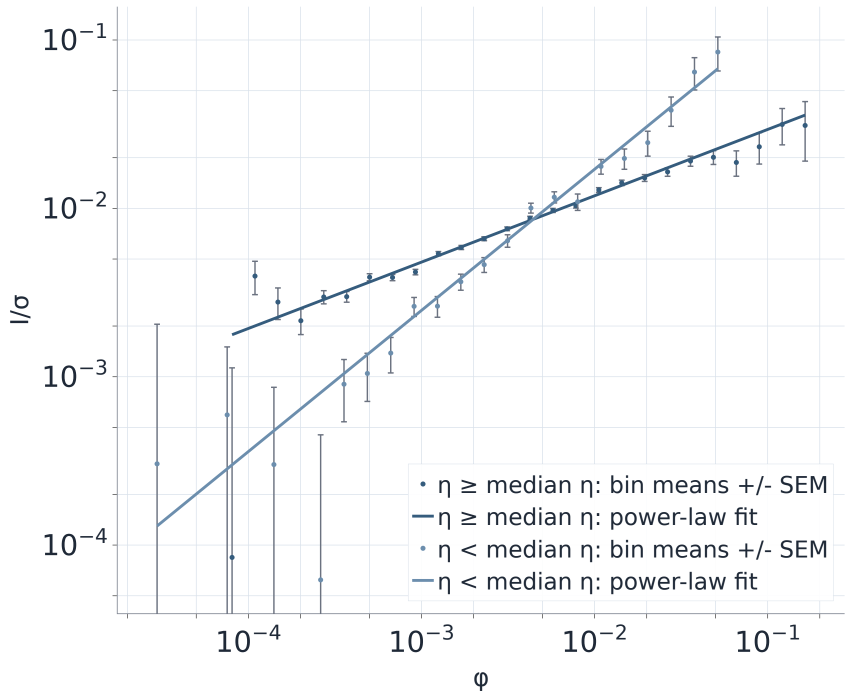
$$\tau = \frac{t - t_i^{\text{start}}}{T_i} \in [0, 1] \qquad (t_i^{\text{start}} \leq t \leq t_i^{\text{end}}), \tag{23}$$

so that $\tau = 0$ corresponds to the first trade and $\tau = 1$ to the last trade. In the aftermath, we continue to track impact on a time window proportional to the execution duration,

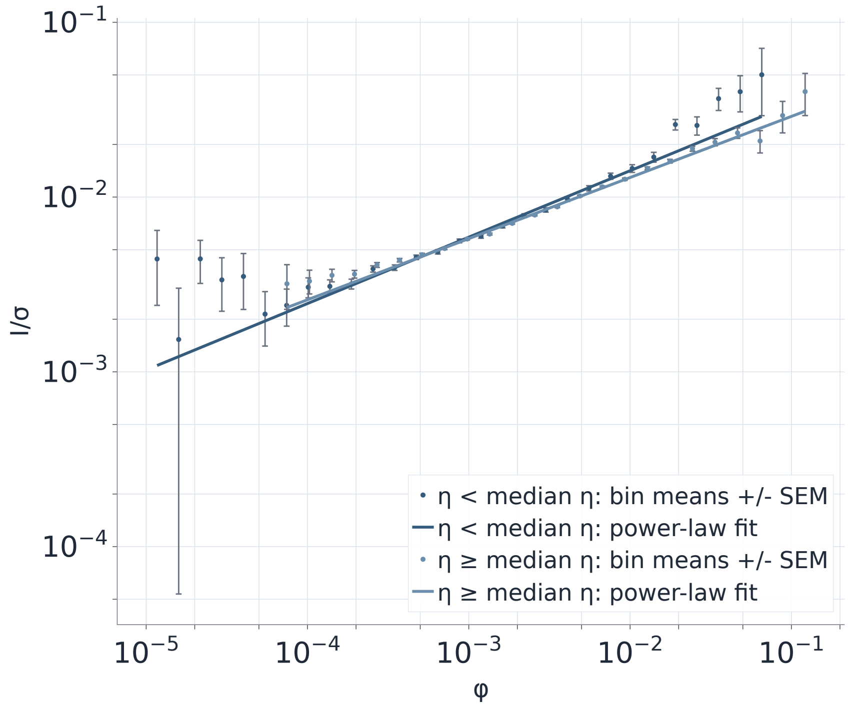
$$\tau = 1 + \frac{t - t_i^{\text{end}}}{T_i} \in (1, 1 + \lambda], \tag{24}$$

for $t_i^{\text{end}} < t \leq t_i^{\text{end}} + \lambda T_i$, where λ has been fixed to 2. For each metaorder we record:

- a partial impact path, i.e. the sequence of $I_i(t)$ evaluated after each child trade during execution ($\tau \in [0, 1]$);
- an aftermath impact path, i.e. $I_i(t)$ sampled at a set of evenly spaced times in the interval $(1, 1 + \lambda]$.



(a) Client



(b) Proprietary

Figure 8: Impact curves conditional on participation rate median, for client and proprietary flow. Each panel shows $\eta < \eta_{\text{median}}$ (green) vs. $\eta \geq \eta_{\text{median}}$ (blue) subsets; markers are bin means $\pm$ SEM and lines are power-law fits.

These two vectors are stored per metaorder and subsequently interpolated onto a common normalised time grid $\{\tau_m\}_{m=0}^{M} \subset [0, 1+\lambda]$. Averaging across metaorders in a given group G (proprietary or client) yields the mean normalised impact path

$$\bar{I}_G(\tau_m) = \frac{1}{N_G(\tau_m)} \sum_{i \in G(\tau_m)} I_i(\tau_m), \tag{25}$$

where $G(\tau_m)$ is the set of metaorders in group G with a defined impact at time τ_m, and $N_G(\tau_m) = |G(\tau_m)|$. The sample used here is the same filtered metaorder set as in the power-law fits (minimum duration, minimum relative size, finite impact).

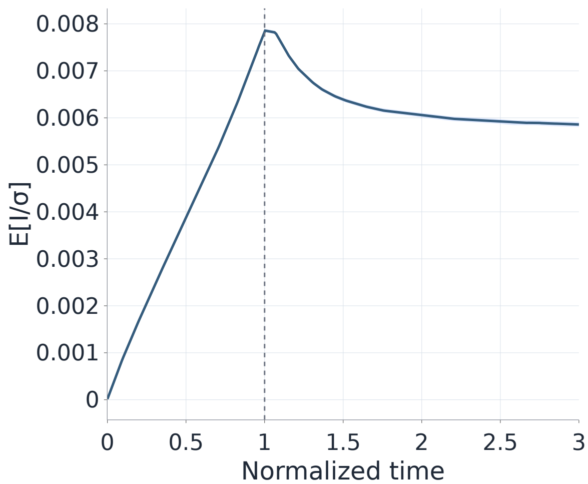
Figure 9 displays the mean paths for proprietary and client (non-proprietary) metaorders, both overall and split by buy/sell. The vertical dashed line marks $\tau = 1$, the end of execution; $\tau > 1$ corresponds to the aftermath window.

For client metaorders, the mean impact increases smoothly over the execution interval and peaks at $\tau \approx 1$. After the end of the execution, the average price relaxes to $\bar{I}(\tau{=}3)/\bar{I}(\tau{=}1) \approx 0.57$, indicating a sizeable transient component. The buy/sell split highlights a pronounced asymmetry for client flow: sell metaorders retain more impact (about 0.67) than buys (about 0.47) at $\tau = 3$ relative to $\tau = 1$; proprietary buy/sell paths are instead very similar (retention about 0.74–0.76).

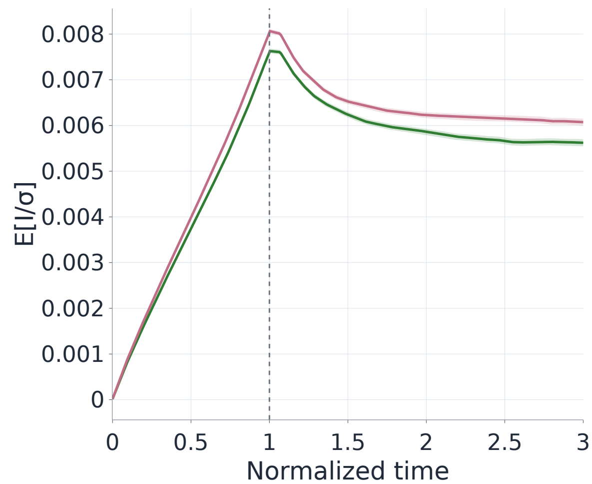
For proprietary metaorders the decay is smaller, since $\bar{I}(\tau{=}3)/\bar{I}(\tau{=}1) \approx 0.75$. We question the null hypothesis of zero differece between the two impact ratios via a bootstrap hypothesis test. We sampled with replacement the days within our dataset for 10000 times, recomputing the same difference over each sample. The results show that there is no sufficient evidence to accept the null hypothesis ($p < 0.001$). This suggests that proprietary metaorders leave a more persistent footprint in prices, whereas client metaorders are associated with a larger transient component that tends to mean-revert. In line with the literature, the persistent component is often interpreted as a proxy for the information content of the metaorder [13]. Overall, the dynamic profiles are in line with the picture of transient but partly permanent impact emerging from empirical studies of metaorders and propagator models: impact accumulates during execution, then relaxes slowly towards a non-zero asymptotic level rather than reverting to zero. In our dataset, the stronger permanence of proprietary impact is consistent with a view in which proprietary trading incorporates more information, while client flow is more liquidity-demanding and hence more prone to subsequent partial reversal. This dynamic perspective complements the static power-law fits of Section 4.2**Pero' l'impatto di picco e' diverso per i due gruppi** and provides additional evidence that market impact and its decay depend sensitively on who trades and how their orders are crowded in time and across investors.

Nationality conditioning provides a useful check on how representative these aggregate curves are. Figure 10 reports the same exercise for the Italian executing-member subsample. The picture is materially different from the aggregate one: the pooled Italian paths are much noisier and, in both proprietary and client flow, the mean signed impact path is negative for most of the horizon. Taken at face value, this means that Italian executing brokers have negative average impact in this subsample, in the sense that prices move on average against the direction of their metaorders after execution starts. The buy/sell split also does not recover the clean, stable pattern seen in the full sample, especially for proprietary Italian metaorders, whose confidence bands are very wide. We therefore read the negative Italian path as a descriptive feature of the sample rather than as conclusive evidence of a distinct structural regime.

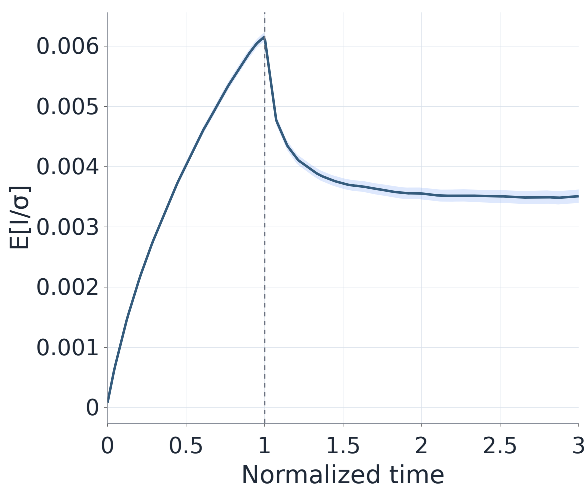
By contrast, the corresponding foreign-member paths are visually indistinguishable from the overall curves in Figure 9. This is exactly what one should expect from the composition results in Section 3.3: foreign brokers account for 99.64% of proprietary metaorders and 90.60% of client metaorders, so the aggregate dynamic-impact estimates are effectively foreign-member estimates.



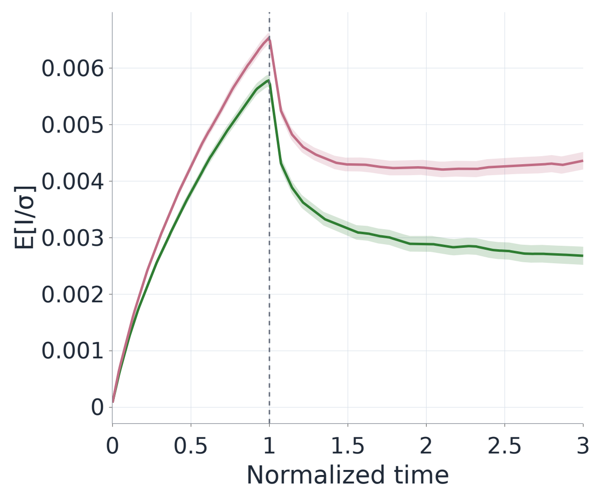
(a) Proprietary: overall



(b) Proprietary: buy vs sell

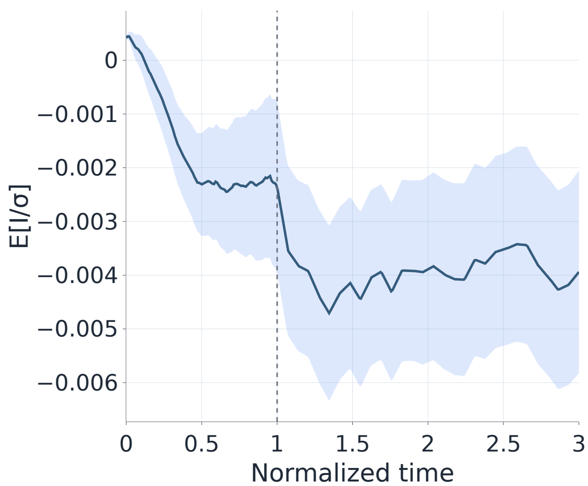


(c) Client: overall

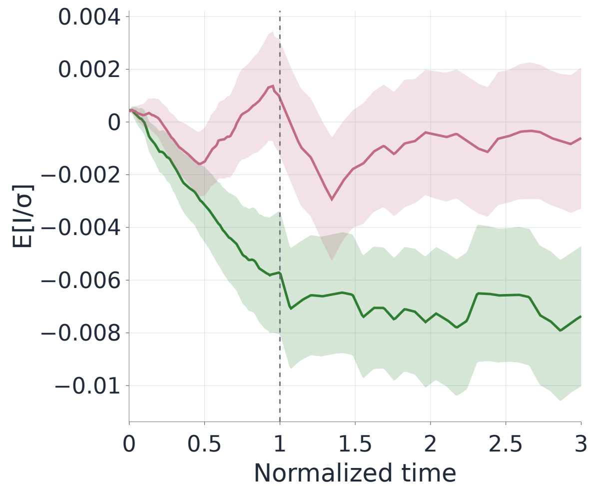


(d) Client: buy vs sell

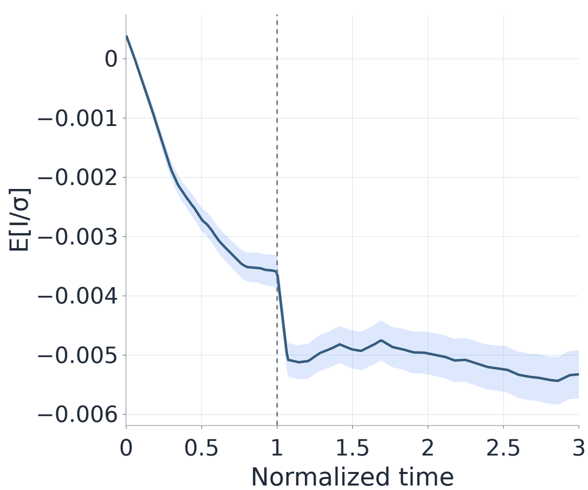
Figure 9: Mean normalised impact paths $\bar{I}_G(\tau)$. Time is normalised so that $\tau = 0$ at the start of the metaorder and $\tau = 1$ at its end; the dashed line marks $\tau = 1$. Shaded regions are ±SEM. Left column: overall paths. Right column: buy vs sell paths (green = buys, red = sells).



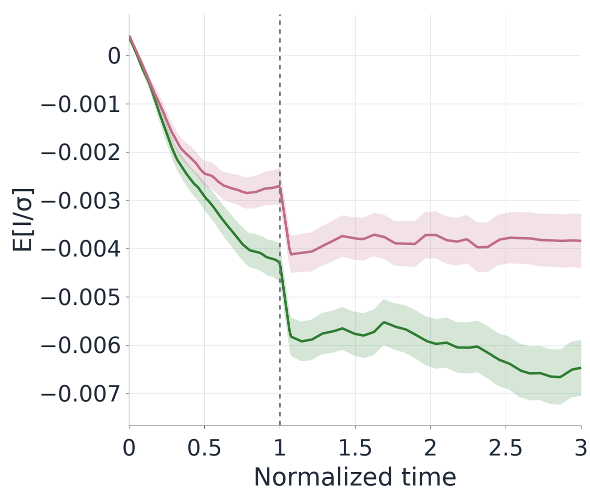
(a) Proprietary: overall



(b) Proprietary: buy vs sell



(c) Client: overall



(d) Client: buy vs sell

Figure 10: Mean normalised impact paths $\bar{I}_G(\tau)$ for the Italian executing-member subsample. The analogous foreign-member paths are omitted because they are visually indistinguishable from the overall curves. Time is normalised so that $\tau = 0$ at the start of the metaorder and $\tau = 1$ at its end; the dashed line marks $\tau = 1$. Shaded regions are ±SEM. Left column: overall paths. Right column: buy vs sell paths (green = buys, red = sells).

5 Crowding and Co-Impact of Proprietary vs Client Flow

We now turn to crowding, namely the tendency of metaorders to align with (or against) the contemporaneous net order flow of other metaorders. Crowding is a key driver of co-impact and execution costs, since metaorders that trade in the same direction as the rest of the market face more adverse price pressure. This connects directly to the institutional co-impact picture of [21].

All crowding measures are based on volume-weighted imbalances constructed from other metaorders in an appropriate reference set.

5.1 Crowding vs all others

We begin from the broadest reference environment. For each instrument k and day d, let $\mathcal{M}_{k,d}$ be the set of all reconstructed metaorders on that stock and date, pooling proprietary and client flow. For a target metaorder $i \in \mathcal{M}_{k,d}$, with size Q_i and direction $\varepsilon_i \in \{-1,+1\}$, we define the leave-one-out all-flow imbalance as

$$\mathrm{imb}_i^{\mathrm{all}}(k,d) = \frac{\displaystyle\sum_{j\in\mathcal{M}_{k,d}\setminus\{i\}} Q_j\varepsilon_j}{\displaystyle\sum_{j\in\mathcal{M}_{k,d}\setminus\{i\}} Q_j}. \tag{26}$$

The statistic is left undefined when the denominator is zero. Positive values mean that, for the day and stock where metaorder i is traded, the rest of the metaorders are buy-dominated, while negative values mean that they are sell-dominated.

Throughout this section we summarize crowding with two related quantities. The first is the correlation between the target direction and the relevant imbalance,

$$r^{G|all} = \mathrm{Corr}\Big(\varepsilon_i, \mathrm{imb}_i^{all}(k,d) \mid i \in G\Big), \qquad G \in \{\mathrm{prop}, \mathrm{client}\}, \tag{27}$$

computed either on the full sample or date by date, i.e.

$$r_d^{G|all} = \mathrm{Corr}\Big(\varepsilon_i, \mathrm{imb}_i^{all}(k,d') \mid i \in G, \quad d'=d\Big), \qquad G \in \{\mathrm{prop}, \mathrm{client}\}$$

Metaorders executed on the same trading day are not independent, as they share market conditions and common shocks. To obtain dependence-aware uncertainty, we construct confidence intervals via a Date-cluster bootstrap: we resample trading days with replacement, recompute the relevant correlation on the resampled panel, and report percentile 95% confidence intervals (1000 replicates).

Averaging $r_d^{G|all}$ over days , we obtain mean daily correlations: $\bar{r}_d^{\mathrm{prop|all}} = 0.041$ with 95% CI $[0.037, 0.046]$ and $\bar{r}_d^{\mathrm{client|all}} = 0.113$ with 95% CI $[0.105, 0.119]$. We observe that client flow shows a stronger alignment with the overall daily imbalance than proprietary flow. This broad benchmark therefore already indicates that client metaorders are more crowded when crowding is measured relative to the aggregate same-stock metaorder environment. The time series of the three day rolling average of $r_d^{G|all}$ is reported in the top panel of Figure 11.

5.2 Within-group and cross-group crowding

Given a group G (proprietary or client), an instrument k, and a day d, let $G_{k,d}$ be the set of metaorders in group G on (k,d). For metaorder $i \in G_{k,d}$, the orderflow imbalance from other metaorders in the same stock and day is defined as

$$\mathrm{imb}_i^{same}(k,d) = \frac{\sum\limits_{j \in G_{k,d} \setminus \{i\}} Q_j \varepsilon_j}{\sum\limits_{j \in G_{k,d} \setminus \{i\}} Q_j}. \tag{28}$$

Analogously to above, the Pearson correlation between metaorder direction and orderflow imbalance is

$$r^{G|same} = \mathrm{Corr}(\varepsilon_i, \mathrm{imb}_i^{same}(k,d) \mid i \in G), \tag{29}$$

computed over all metaorders in the group, all stocks and all days. We also compute daily correlations r_d by restricting to metaorders on a single date d and evaluating

$$r_d^{G|same} = \mathrm{Corr}(\varepsilon_i, \mathrm{imb}_i^{k,d'} \mid i \in G, \quad d' = d) \tag{30}$$

computed for days with at least 100 metaorders and averaged across stocks. The time series of the three day rolling average of $r_d^{G|same}$ is plotted for proprietary and client flows separately in the middle panel of Figure 11.

In our dataset, within-group orderflow imbalance statistics are:

- Proprietary flow.
  - Global correlation: $r^{prop|same} = 0.108$ with 95% CI $[0.102, 0.113]$, $n = 588{,}334$ metaorders.
  - Mean daily correlation: $\bar{r}_d^{prop|same} = 0.081$ with 95% CI $[0.077, 0.086]$ over 251 days.
- Client flow.
  - Global correlation: $r^{client|same} = 0.187$ with 95% CI $[0.180, 0.196]$, $n = 255{,}535$ metaorders.
  - Mean daily correlation: $\bar{r}_d^{client|same} = 0.169$ with 95% CI $[0.162, 0.177]$ over 251 days.

Thus both client and proprietary metaorders tend to trade with the local daily imbalance (positive crowding/herding), although client metaorders show a statistically significant larger correlation coefficient.

Another useful perspective is to consider the alignment of a metaorder with the imbalance of all the metaorders in the other group. To study this cross-group crowding, we construct an environment imbalance based solely on the other group. For example, to ask whether proprietary metaorders trade with or against the imbalance generated by clients, we proceed as follows.

Fix a source group G_{src} (e.g. clients) and a target group G_{tgt} (e.g. proprietary). For each instrument k and day d, let $G_{k,d}^{\mathrm{src}}$ be the set of metaorders in the source group on (k,d). Define the environment imbalance for that (k,d) as

$$\mathrm{imb}^{\mathrm{env}}(k,d) = \frac{\sum\limits_{j \in G_{k,d}^{\mathrm{src}}} Q_j \varepsilon_j}{\sum\limits_{j \in G_{k,d}^{\mathrm{src}}} Q_j}. \tag{31}$$

This is then attached to each target-group metaorder i in the same (k,d) as $\mathrm{imb}_i^{\mathrm{env}}$.

We compute daily correlations

$$r_d^{\text{prop}|\text{client}} = \text{Corr}\big(\varepsilon_i, \text{imb}_i^{\text{env}}(k, d') \,\big|\, i \in prop, \quad d' = d\big), \tag{32}$$

$$r_d^{\text{client}|\text{prop}} = \text{Corr}\big(\varepsilon_i, \text{imb}_i^{\text{env}}(k, d') \,\big|\, i \in client, \quad d' = d\big), \tag{33}$$

where the environment is built from client or proprietary flow, respectively, and the results are averaged across all stocks.

Empirically we find

- Global correlations:
  - $\text{Corr}(\varepsilon_{\text{prop}}, \text{imb}_{\text{client}}^{\text{env}}) = -0.023$ with 95% CI $[-0.028, -0.018]$.
  - $\text{Corr}(\varepsilon_{\text{client}}, \text{imb}_{\text{prop}}^{\text{env}}) = -0.016$ with 95% CI $[-0.025, -0.007]$.
- Mean daily correlations (days with $n \geq 100$):
  - $\bar{r}_d^{\text{prop}|\text{client}} = -0.018$ with 95% CI $[-0.023, -0.014]$,
  - $\bar{r}_d^{\text{client}|\text{prop}} = -0.003$ with 95% CI $[-0.010, 0.004]$.

The prop | client daily correlation is slightly negative, indicating a (very) mild tendency for proprietary group to trade against the imbalance produced by the client group, which is reminiscent of the way different institutional styles interact in [21]. The client | prop correlation is close to zero with a confidence interval that includes zero, suggesting no clear alignment or anti-alignment of client metaorders with the proprietary environment. The time series of the three day rolling average of these cross-group correlations is shown in the bottom panel of Figure 11.

5.3 Crowding conditioned on participation rate

The previously defined execution participation rate η_i provides a measure for the "aggressiveness" of a metaorder, i.e. the fraction of the contemporaneous traded volume absorbed during its own execution window. We use η_i as an execution-state variable and ask whether crowding becomes stronger at higher participation.

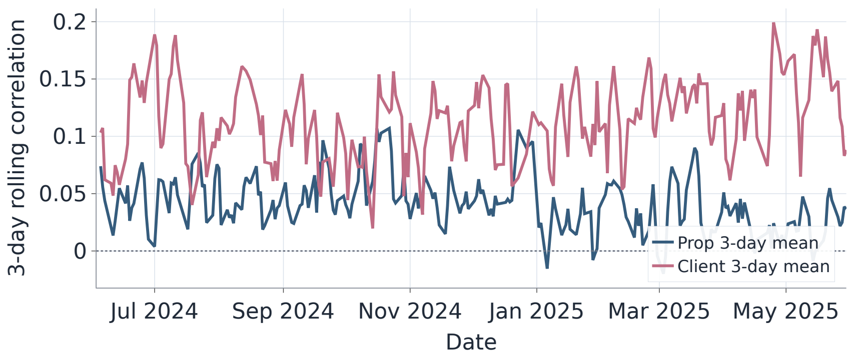
For each group (proprietary and client) separately, we partition metaorders into deciles of η_i using pooled quantiles (common bin edges across groups). Within each bin b we compute two statistics based on an imbalance proxy imb_i: **Da riscrivere con la nuova notazione**

$$A_b = \mathbb{E}[\varepsilon_i \,\text{imb}_i \mid \eta_i \in b], \tag{34}$$

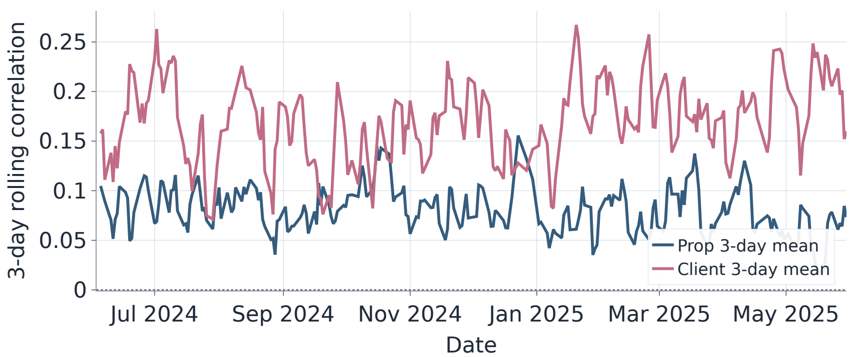
$$U_b = \mathbb{E}[|\text{imb}_i| \mid \eta_i \in b]. \tag{35}$$

where imb_i is either the within-group leave-one-out imbalance $\text{imb}_i^{k,d}$ from Eq. (28) or the cross-group environment imbalance $\text{imb}_i^{\text{env}}$ defined above. Confidence bands are obtained via a block bootstrap that resamples trading days with replacement (1000 replicates), which accounts for cross-sectional dependence within the same date. The corresponding top-minus-bottom decile contrasts, which summarize the magnitude of these participation-rate gradients, are reported in Table 6.

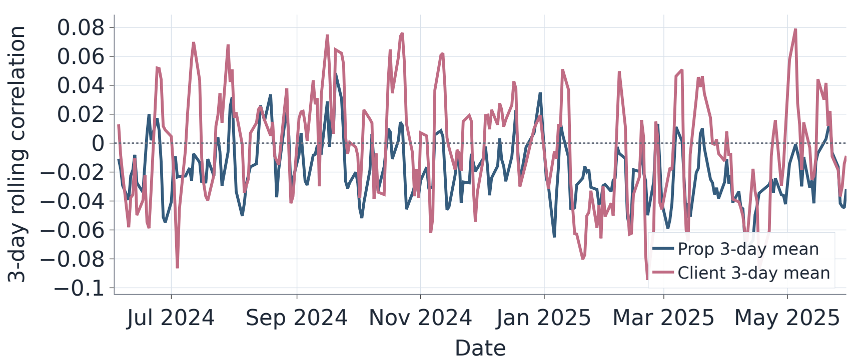
Within-group conditioning. Figure 12 shows that both groups become more crowded as η increases. The effect is monotone and much stronger for client flow: high-participation client metaorders are strongly aligned with the contemporaneous within-client imbalance, while low-η metaorders are close to neutral. Proprietary flow exhibits a milder but statistically significant increase. In addition, U_b increases with η for both groups, indicating that high-participation executions tend to occur on (k, d) pairs where the rest of the same-group flow is more one-sided.



(a) All other metaorders.



(b) Within-group imbalance.



(c) Cross-group environment.

Figure 11: Crowding rolling time series. Each panel reports the 3-day rolling mean of the relevant daily direction–imbalance correlation. Top: each target group against all other metaorders. Middle: within-group crowding. Bottom: cross-group crowding.

Imbalance proxy	Group	ΔA	ΔU
$\mathrm{imb}_i^{k,d}$	Proprietary	0.0296 [0.0237, 0.0359]	0.0280 [0.0203, 0.0360]
$\mathrm{imb}_i^{k,d}$	Client	0.1890 [0.1680, 0.2087]	0.0253 [0.0164, 0.0346]
$\mathrm{imb}_i^{\mathrm{env}}$	Proprietary	−0.0057 [−0.0136, 0.0028]	0.0453 [0.0360, 0.0547]
$\mathrm{imb}_i^{\mathrm{env}}$	Client	−0.0447 [−0.0551, −0.0358]	0.0244 [0.0183, 0.0305]

Table 6: Change in crowding measures between the top and bottom participation-rate deciles. We report ΔA and ΔU (top minus bottom decile) with 95% confidence intervals from a day-cluster bootstrap (1000 replicates).

Cross-group conditioning. Figure 13 reveals a different pattern across groups. While the magnitude of the other group's imbalance increases with η for both proprietary and client metaorders, the alignment statistics diverge. Proprietary metaorders are weakly contrarian to the client environment across η with no pronounced trend, whereas client metaorders become increasingly contrarian to the proprietary environment at high η. This behaviour is consistent with an intermediation/risk-transfer picture in which aggressive client demand is partly offset by proprietary flow.

5.4 Member level crowding

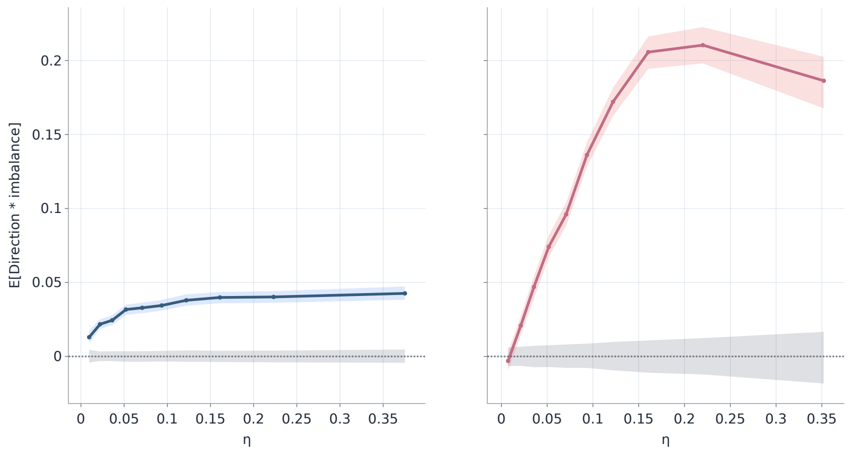
The final measure of interest—particularly from a regulatory perspective—is the correlation between a member's proprietary trading direction and the order-flow imbalance generated by the clients whose metaorders were executed through that member. This indicator can be interpreted as evidence that the member either provides liquidity to its clients' metaorders (negative correlation) or trades in the same direction as the flow, potentially amplifying the market impact of its clients' trades (positive correlation).

For a given member m and date d, let $\mathcal{C}_{m,d}$ denote the set of client metaorders executed by member m on day d, aggregated across all ISINs. We define the member-level client imbalance as

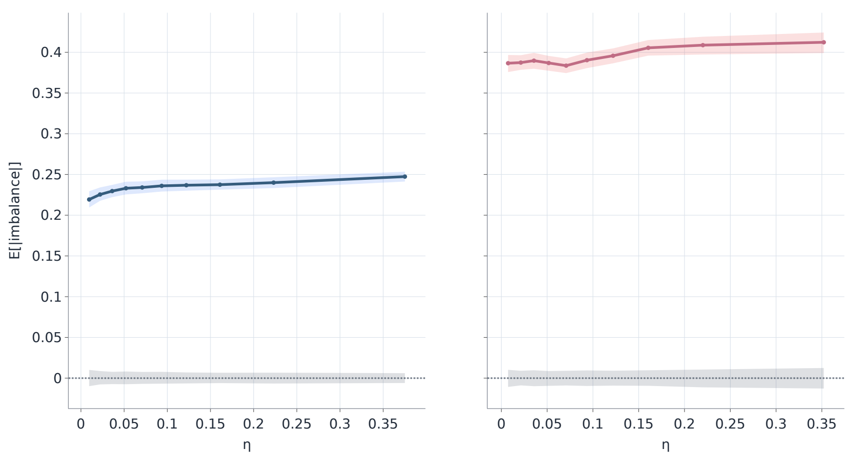
$$\mathrm{imb}_{m,d} = \frac{\sum_{j\in\mathcal{C}_{m,d}} Q_j \epsilon_j}{\sum_{j\in\mathcal{C}_{m,d}} Q_j}, \tag{36}$$

where Q_j and ϵ_j denote, respectively, the size and direction of client metaorder j. The statistic $\mathrm{imb}_{m,d}$ is considered valid only when both the member and its associated client cohort initiate a sufficient number of metaorders during day d. Using members with at least 30 proprietary and 30 client metaorders over the sample, there are 10 such members (out of 45), and the mean per-member correlation between proprietary direction and client imbalance is shown in Figure 14. As we can see, most members have correlations close to zero, but a few exhibit stronger positive or negative correlations with peak values near +0.4 suggesting that the relationship between proprietary trading and client flow is quite heterogeneous across members and not trivial.

Discutere se lasciarlo: forse lo metterei in appendice e metterei qui una frase. Non capisco bene nella figura le colonne completamente bianche.... A more informative analysis focuses on how this correlation evolves over time. Periods displaying larger absolute values may signal episodes of front running (when $\mathrm{imb}_{m,d} > 0$) or systematic liquidity provision (when $\mathrm{imb}_{m,d} < 0$). Figure 15 reports a heatmap of the correlation computed over non-overlapping 3-day windows. The white vertical gaps are not zero-correlation periods: they correspond to windows in which no displayed member has enough proprietary and same-member client activity to pass the minimum-count filter, so the member-window correlation is undefined and the cell is omitted. As illustrated, most member-window blocks hover near zero, but a handful of windows show stronger

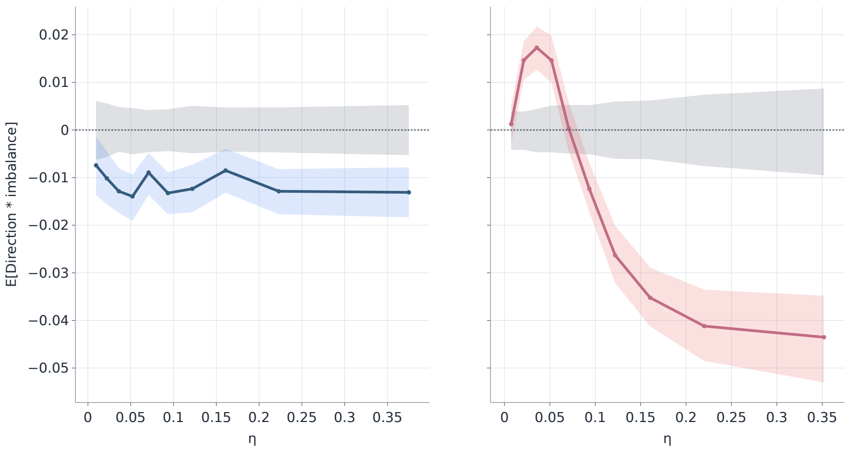


(a) Mean alignment $\mathbb{E}[\varepsilon_i \, \mathrm{imb}_i^{k,d} \mid \eta]$.

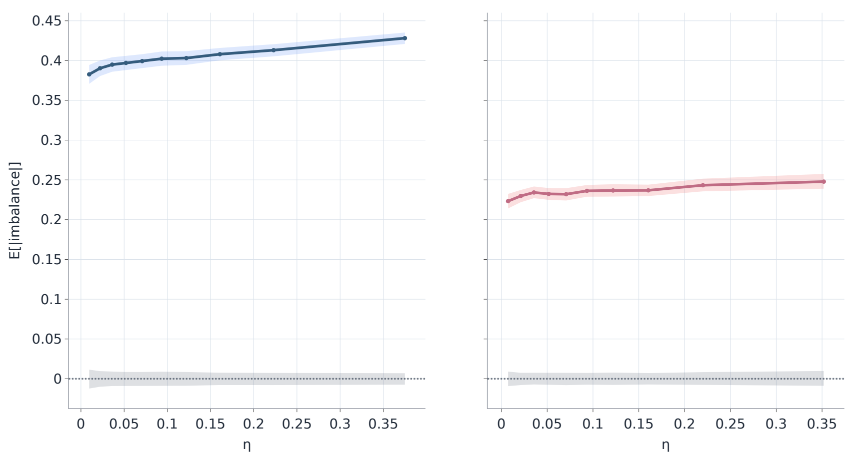


(b) Mean absolute imbalance $\mathbb{E}[|\mathrm{imb}_i^{k,d}| \mid \eta]$.

Figure 12: Within-group crowding conditioned on participation rate deciles. Each plot has two panels (left: proprietary, right: client). Shaded regions are 95% day-cluster bootstrap confidence bands.



(a) Mean alignment $\mathbb{E}[\varepsilon_i \, \mathrm{imb}_i^{\mathrm{env}} \mid \eta]$.



(b) Mean absolute imbalance $\mathbb{E}[|\mathrm{imb}_i^{\mathrm{env}}| \mid \eta]$.

Figure 13: Cross-group crowding conditioned on participation rate deciles. The environment imbalance $\mathrm{imb}_i^{\mathrm{env}}$ is constructed from the other group on the same (k, d). Each plot has two panels (left: proprietary, right: client); shaded regions are 95% day-cluster bootstrap confidence bands.

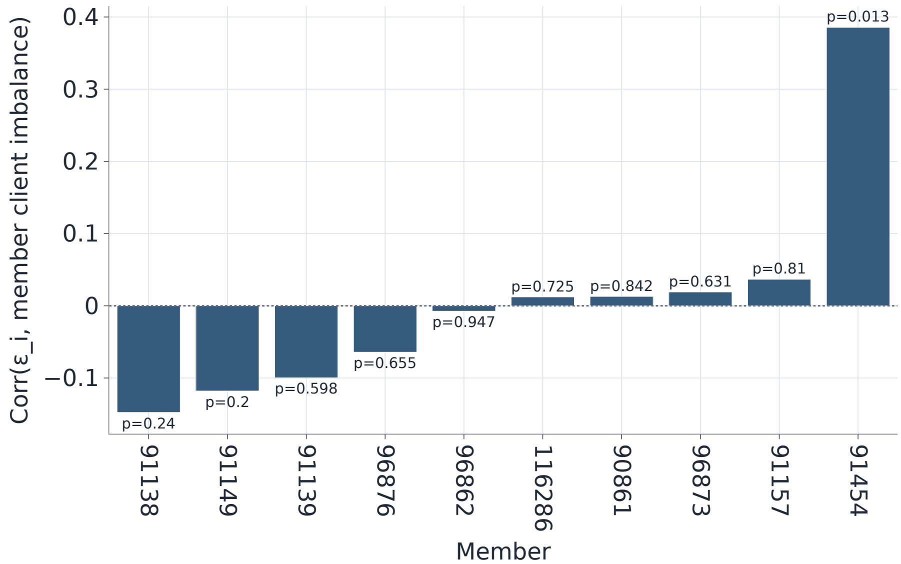


Figure 14: Per-member correlations between proprietary trading direction and member-level client imbalance (members with at least 30 proprietary and 30 client metaorders; horizontal line marks zero).

alignment or opposition (often in small-sample windows), suggesting that prop–client co-movement is episodic and member-dependent rather than uniformly persistent.

5.5 Does crowding explain the higher impact of proprietary flow?

The previous subsections show that crowding is economically meaningful for both groups and, when measured at the daily stock level, often stronger for client flow. We now return to the question raised by the impact curves in Section 4.2: why do proprietary metaorders have larger impact over the empirically relevant range of Q/V?. In this section, we present several analysis that, step by step, give us more intuition regarding theThe answer source of the impact differences.

We first use a daily, stock-level notion of crowding as done in the previous subsection. To place proprietary and client metaorders in the same reference environment, we use the all-vs-all imbalance $\mathrm{imb}_i^{\mathrm{all}}$ and the aligned-crowding variable

$$c_i = \varepsilon_i \, \mathrm{imb}_i^{\mathrm{all}}. \tag{37}$$

As usual, positive c_i means that metaorder i trades in the same direction as the aggregate same-stock same-day imbalance generated by all other metaorders.

We then sort proprietary and client metaorders separately into terciles of c_i and re-estimate the standard power-law model $I = Y\phi^{\gamma}$ **e il segno e la volatilita' che fine hanno fatto?** E' all'interno di I (equazione 16) within each tercile. Table 7 reports the difference between fitted impact in the highest and lowest crowding terciles at two benchmark sizes, while Figure 16 shows the overall curves. The difference is positive for both groups: metaorders that trade with the same-day imbalance have larger fitted impact. The effect, however, is larger for clients and statistical significant as shown in the last row of Table 7. **Di nuovo non capisco: sono i clienti a essere piu' crowded, no?**Si, ma finora non avevamo ancora analizzato la dipendenza della power law dalla

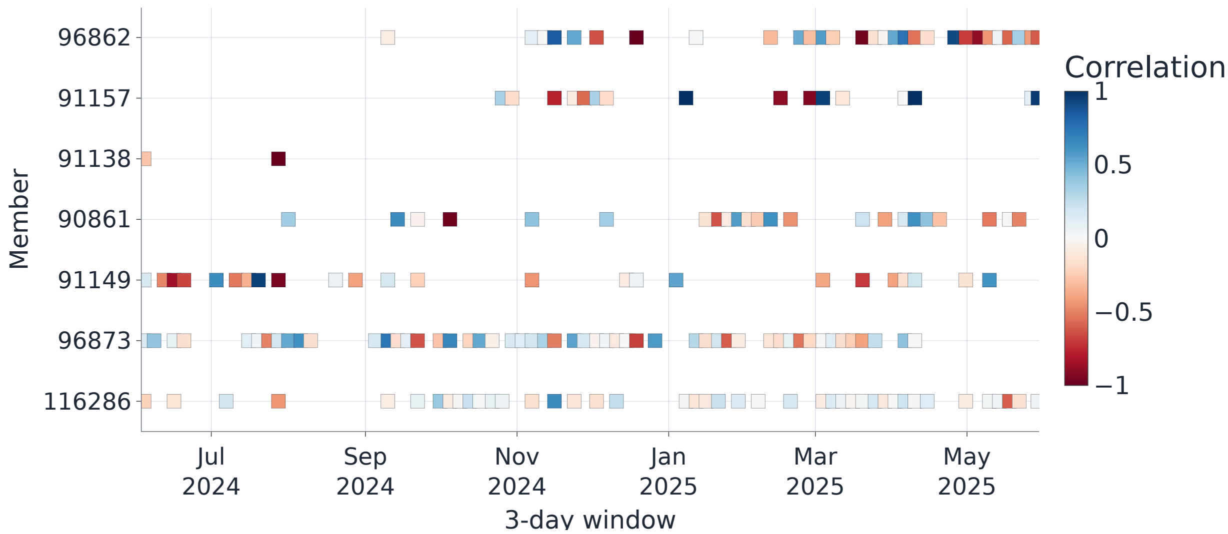


Figure 15: Sparse heatmap of the correlation between proprietary metaorder directions and the same-member client order-flow imbalance, computed over non-overlapping 3-day windows. Only member-window cells passing the minimum-count filter are shown: both the proprietary desk and its client cohort must execute at least 5 metaorders in that window. Members with no displayed cells are omitted. Colors indicate the sign and magnitude of $\mathrm{corr}(\epsilon_i, \mathrm{imb}_{m,d})$, with blue (red) denoting positive (negative) association. The figure highlights several members exhibiting episodic periods of stronger alignment or opposition.

metrica del crowding. Questi risultati ci dicono che, in aggiunta al crowding minore dei proprietari, non c'è nemmeno un aumento maggiore dell'impatto all'aumentare del crowding. Ho modificato la frase che era in effetti poco chiara.. These results suggest that proprietary metaorders not only exhibit lower average crowding but also demonstrate a reduced sensitivity of market impact to the crowding variable.

FAB:FIN QUI

However, because aligned crowding and participation rate are themselves positively related (see Section 5.3), especially for clients, Figure 17 repeats the comparison conditioned on two levels of η (smaller and larger then the median). The crowding premium remains visible, but becomes more heterogeneous, indicating that crowding and execution aggressiveness interact rather than entering independently.

As a second daily-crowding check, we extend the regression analysis of normalized impact versus metaorders relative size with the imbalance magnitude and execution state. Let $v_i \equiv V_i^{\mathrm{window}}/V_{d(i)} = \phi_i/\eta_i$ be the metaorder duration in volume time. We form common pooled bins in $(\eta_i, v_i, |\mathrm{imb}_i^{\mathrm{all}}|)$, compute the mean signed impact $\bar{I}_b$ in each retained positive-mean cell b, and estimate

$$\log \bar{I}_b = a + \beta_\eta \log \eta_b + \beta_v \log v_b + \beta_c |\mathrm{imb}_b^{\mathrm{all}}| + \beta_p \mathbf{1}\{G_b = \mathrm{prop}\} + \xi. \tag{38}$$

As in the standard impact fits, the regression is estimated by WLS, with weights based on the delta-method variance of $\log \bar{I}_b$ computed from the cell-level SEM. Finallu, ξ denotes the residual. Table 8 shows the estimated coefficients of the fit. We observe that $|\mathrm{imb}^{\mathrm{all}}|$ enters positively, but the proprietary dummy remains large and significant. The estimate $\hat{\beta}_p = 0.173$ implies roughly 19% higher mean impact for proprietary cells with the same participation rate, execution-window

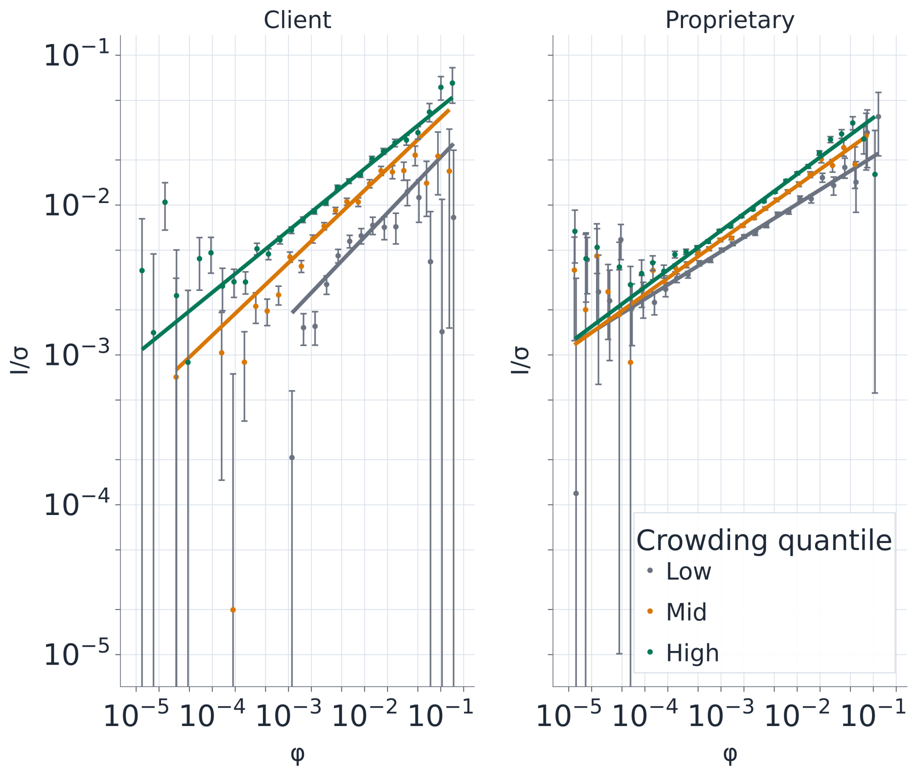


Figure 16: Impact curves by crowding tercile.

volume share, and daily crowding magnitude. Daily imbalance is therefore relevant for impact, but it does not absorb the proprietary impact premium.

The daily-imbalance tests therefore point to a more local question. The impact curves measure the price response during and after a metaorder; a relevant crowding channel may therefore be the flow that is alive during the target metaorder's execution, not only the same-day aggregate imbalance. For target metaorder i and another metaorder j in the same $(\mathrm{ISIN}, \mathrm{Date})$, we therefore define the active-overlap weight

$$\omega_{ij} = \frac{\left[\min(t_i^{\mathrm{end}}, t_j^{\mathrm{end}}) - \max(t_i^{\mathrm{start}}, t_j^{\mathrm{start}})\right]_+}{t_i^{\mathrm{end}} - t_i^{\mathrm{start}}}, \tag{39}$$

with the convention that a zero-duration target has $\omega_{ij} = 1$ if $t_j^{\mathrm{start}} \le t_i^{\mathrm{start}} \le t_j^{\mathrm{end}}$ and zero otherwise. We then compute leave-one-out active-overlap features, including the time-weighted number of active metaorders, gross active volume normalized by the target volume, same-sign and opposite-sign active volumes, and the target-signed active imbalance. Writing $\mathcal{A}_i$ for all other metaorders in the same $(\mathrm{ISIN}, \mathrm{Date})$ as target i, these quantities are:

- the active count $N_i^{\mathrm{act}} = \sum_{j \in \mathcal{A}_i} \omega_{ij}$
- the gross active volume normalized by target size $G_i^{\mathrm{act}}/Q_i = \sum_{j \in \mathcal{A}_i} \omega_{ij} Q_j / Q_i$
- the same-sign and opposite-sign active volumes $S_i^{\mathrm{act}} = \sum_{j:\varepsilon_j = \varepsilon_i} \omega_{ij} Q_j$ and $O_i^{\mathrm{act}} = \sum_{j:\varepsilon_j = -\varepsilon_i} \omega_{ij} Q_j$
- the target-signed active imbalance $B_i^{\mathrm{act}} = (S_i^{\mathrm{act}} - O_i^{\mathrm{act}})/(S_i^{\mathrm{act}} + O_i^{\mathrm{act}})$.

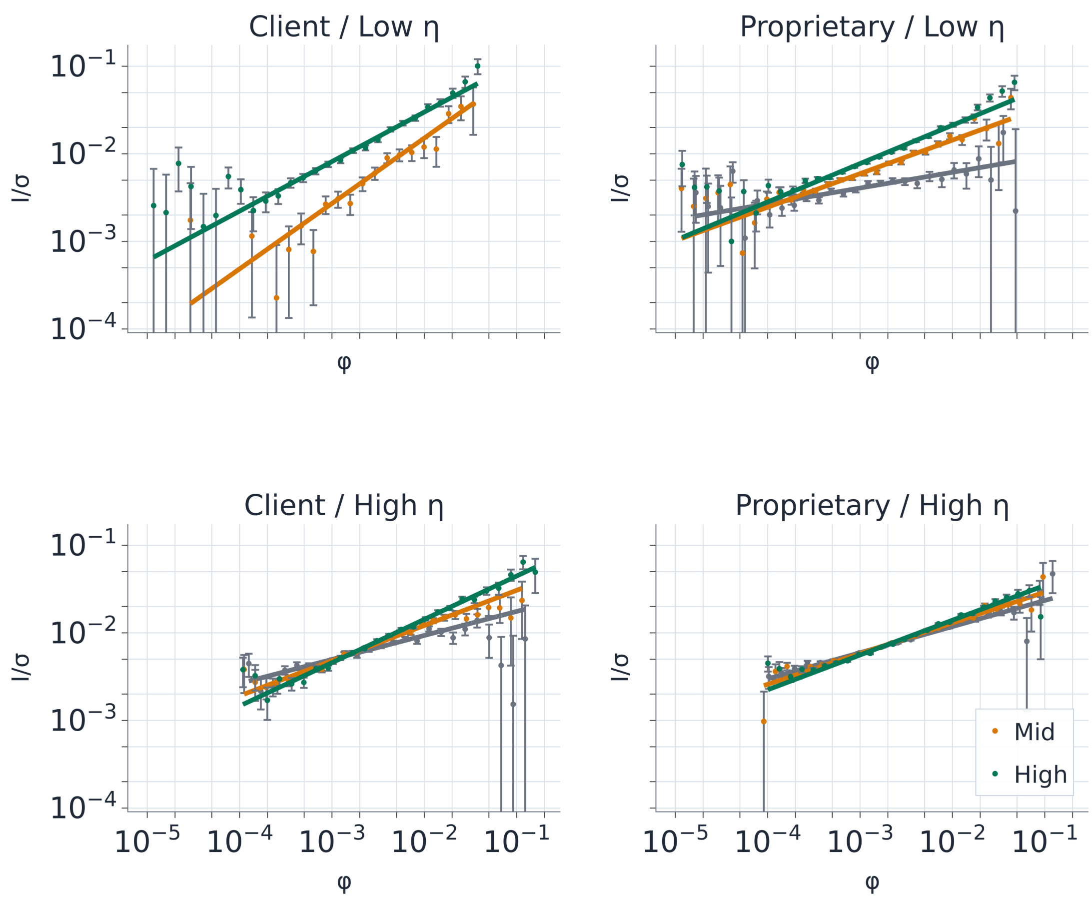


Figure 17: Impact curves by crowding tercile across two participation rate bins.

Group	$\phi = 10^{-3}$ High − low	95% CI	$\phi = 10^{-2}$ High − low	95% CI
Client	0.00496	[0.00448, 0.00579]	0.01135	[0.00968, 0.01285]
Proprietary	0.00186	[0.00155, 0.00213]	0.00592	[0.00502, 0.00684]
Prop. − client	-0.00310	[-0.00377, -0.00272]	-0.00543	[-0.00644, -0.00424]

Table 7: Difference between fitted impacts in the highest and lowest terciles of aligned crowding $c_i = \varepsilon_i \,\mathrm{imb}_i^{\mathrm{all}}$. Positive values indicate that more crowded metaorders have higher fitted impact at the benchmark size ϕ. Confidence intervals are obtained from a day-cluster bootstrap with 200 replicates.

Regressor	Estimate	Bootstrap 95% CI
$\log \eta$	0.227	[0.216, 0.238]
$\log v$	0.265	[0.258, 0.285]
$\lvert\mathrm{imb}^{\mathrm{all}}\rvert$	0.084	[0.019, 0.142]
$\mathbf{1}\{G = \mathrm{prop}\}$	0.173	[0.115, 0.221]

Table 8: Pooled cell-level regression of $\log \bar{I}_b$ on execution-state controls and crowding. Cells are formed using common bins in $(\eta, v, \lvert\mathrm{imb}^{\mathrm{all}}\rvert)$ across proprietary and client flow, with $v = \phi/\eta = V^{\mathrm{window}}/V_d$. The reported regression uses 111 retained cells out of 176 non-empty pooled cells. Bootstrap intervals are obtained from a day-cluster bootstrap with 200 replicates.

Table 9 summarizes the resulting features. Proprietary targets are not more overlapped in count: the mean time-weighted active count is 5.36 for proprietary metaorders and 5.57 for client metaorders. Gross active volume normalized by target size is also not higher for proprietary targets. The statistic that distinguishes the groups is directional: target-signed active imbalance is positive for proprietary metaorders and close to zero for clients. Thus the active-overlap evidence refines the daily-crowding evidence. Proprietary metaorders are not executed in a mechanically denser local daily environment, but the overlapping flow they face is more often aligned with their own direction.

We quantify this channel by estimating three controlled versions of the log-binned WLS impact curves by adding additional regressors each time. The baseline WLS specification includes the proprietary dummy and $\log(Q/V)$; we then add the logs of cell-mean participation and volume duration. Finally, we add positive active-overlap components: the log of cell-mean active count, same-sign active overlap normalized by target size, and opposite-sign active overlap normalized by target size. Table 10 reports the proprietary log-coefficient.

This exercise answers the question closest to the original impact-curve evidence. Participation rate and duration alone do not reduce the proprietary log-premium: W0 and W1 give similar coefficients, about 0.18, or a 20% multiplicative premium. Once active-overlap summaries are added on the same 51 Q/V curve cells, the coefficient falls to 0.14 and the confidence interval includes zero. This shows that, in the same log-binned WLS language used in 4.2, active directional overlap is the control that most weakens the proprietary premium.

Putting these pieces together, crowding is relevant for impact, but the channel is not simply the number of simultaneous metaorders. Daily crowding is stronger for clients, and active overlap in counts and gross volume is not higher for proprietary targets. What distinguishes proprietary executions is the sign of the active environment. Active directional overlap explains part of the proprietary coefficient at the metaorder level and materially weakens it in the power-law-style binned WLS. We therefore interpret the higher proprietary impact as partly associated with execution in more directionally aligned local environments, while recognizing that the residual magnitude depends on the aggregation level and controls used.

Active-overlap feature	Client mean	Client median	Prop. mean	Prop. median
Time-weighted active count	5.57	5.40	5.36	5.15
Gross active volume / target Q	36.66	10.90	25.55	11.14
Target-signed active imbalance	0.000	0.001	0.026	0.038
Same-sign active volume	177,728	33,959	172,275	32,098
Opposite-sign active volume	183,110	33,344	153,453	28,795

Table 9: Active-overlap features computed over true execution intervals within the same ISIN and date. The target itself is excluded. The active imbalance is signed by the target direction, so positive values indicate that the overlapping flow is aligned with the target metaorder.

Model	Controls	$\hat{\beta}_{\text{prop}}$	$\exp(\hat{\beta}) - 1$	95% CI	Cells	N
W0	$\log(Q/V)$	0.179	19.6%	[0.114, 0.245]	51	833,795
W1	W0 plus $\log(\bar{\eta})$ and $\log(\bar{v})$	0.182	20.0%	[0.072, 0.292]	51	833,795
W2	W1 plus positive active-overlap controls	0.140	15.0%	[-0.065, 0.344]	51	833,795

Table 10: Power-law-style binned log-WLS regressions. The dependent variable is the log mean impact in group-by-Q/V cells. W2 adds the log of positive cell-mean active-overlap controls to the binned WLS specification.

5.6 Active member-level prop–client alignment

The active-overlap results above suggest that the flow alive during the execution interval is the relevant local environment. We now return to the member-level question of Section 5.4: are there members whose proprietary metaorders are systematically aligned, either positively or negatively, with the flow of their own clients? The difference with the member-day statistic is that the client environment is now restricted to active overlapping client metaorders.

For a proprietary metaorder i executed by member m, let $\mathcal{C}^{\text{act}}_{i,m}$ be the set of client metaorders executed through the same member, on the same ISIN and date, that overlap in calendar time with i. Using the active-overlap weights ω_{ij} defined above, the active client imbalance faced by i is

$$\text{imb}^{\text{act}}_{i,m} = \frac{\sum_{j\in\mathcal{C}^{\text{act}}_{i,m}} \omega_{ij} Q_j \epsilon_j}{\sum_{j\in\mathcal{C}^{\text{act}}_{i,m}} \omega_{ij} Q_j}, \tag{40}$$

and is left undefined when the denominator is zero. This is the active, same-ISIN analogue of $\text{imb}_{m,d}$ in Section 5.4. The member-specific statistic is

$$r^{\text{act}}_m = \text{Corr}\left(\epsilon^{\text{prop}}_i, \text{imb}^{\text{act}}_{i,m} \,\middle|\, \text{Member}(i) = m\right). \tag{41}$$

Positive values mean that the proprietary metaorders of member m tend to have the same sign as the active imbalance generated by its own clients; negative values indicate systematic opposition. We also compute the signed alignment $a_i = \epsilon^{\text{prop}}_i \text{imb}^{\text{act}}_{i,m}$ as a descriptive quantity. The correlation in (41) is more stringent than the average of a_i, because it removes unconditional buy/sell biases of the proprietary desk and of the client environment.

The local restriction is severe. Out of 588,334 proprietary metaorders, only 2,320 have a same-member, same-ISIN client metaorder actively overlapping in time. Table 11 reports pooled benchmark correlations. The baseline active-overlap correlation is positive but small, $r^{\text{act}} = 0.036$, with a Date-cluster bootstrap confidence interval including zero. Splitting the client environment by timing gives the same qualitative message: the correlation is 0.051 when the client metaorder is already active at the proprietary start and 0.037 when the client metaorder starts during the proprietary

Environment	Timing bucket	Correlation	95% CI	Valid targets	Members
same ISIN	all active	0.036	[−0.032, 0.104]	2,320	12
same ISIN	active at prop start	0.051	[−0.034, 0.137]	1,446	11
same ISIN	starts during prop	0.037	[−0.038, 0.105]	1,225	12
all ISINs	all active	0.011	[−0.055, 0.077]	20,636	15

Table 11: Pooled active member-level prop–client correlations. The main environment is same-member, same-ISIN active client flow. The all-ISIN row is a robustness check that pools active same-member client metaorders across instruments. Confidence intervals are Date-cluster bootstrap percentile intervals.

Member	r_m^{act}	95% CI	Valid targets	Dates
91138	-0.019	[−0.339, 0.265]	50	35
91149	0.177	[−0.009, 0.388]	195	104
91157	-0.006	[−0.117, 0.084]	946	68
96862	0.431	[0.249, 0.614]	89	61
116286	0.033	[−0.075, 0.156]	981	162

Table 12: Per-member active prop–client correlations in the main same-ISIN, all-active environment. Members are shown when they have at least 30 valid proprietary targets with an active same-member client environment.

interval. Thus the pooled evidence does not point to a uniform member-level tendency to align with own-client flow.

The pooled statistic can mask heterogeneity across members, which is the main object of this exercise. We therefore compute (41) separately by member and report members with at least 30 valid proprietary targets in the relevant environment. In the baseline same-ISIN, all-active specification, five members pass this threshold. Table 12 shows that most of them have correlations close to zero, but member 96862 stands out with $r_m^{\text{act}} = 0.431$ and a confidence interval $[0.249, 0.614]$. Member 91149 is also positive, although its baseline interval barely includes zero. We do not observe a comparable strongly negative same-ISIN member in this specification.

The timing decomposition clarifies where the strongest member-specific signals come from. For member 96862, the correlation remains positive when the client metaorder is already active at the proprietary start: $r_m^{\text{act}} = 0.382$ with 95% CI $[0.059, 0.685]$ over 34 valid targets. The same member also has a large correlation in the "starts during prop" bucket (0.501, 95% CI $[0.273, 0.719]$, 62 targets). Member 91149 is positive in the latter bucket as well (0.269, 95% CI $[0.079, 0.451]$, 99 targets). The first of these results is the most directly connected to alignment with already active client flow; the latter two indicate local co-movement, but do not establish that proprietary trading follows client flow, since the client metaorders start after the proprietary interval has begun.

To make the strongest member-level correlations visually interpretable, we also aggregate the target-level quantities into non-overlapping 5-trading-day windows for the two members with the largest positive global correlations. For member m and window w, we plot the proprietary target-direction imbalance and the average active client imbalance,

$$P_{m,w} = n_{m,w}^{-1} \sum_i \epsilon_i^{\text{prop}}, \qquad C_{m,w} = n_{m,w}^{-1} \sum_i \text{imb}_{i,m}^{\text{act}},$$

where the sums run over valid proprietary targets in that member-window. Unlike the discarded member-window heatmap, this diagnostic imposes no minimum count filter at the window level; sparse windows are simply reflected in the number of targets reported in the figure metadata. We therefore use this panel only as a descriptive co-movement diagnostic; the inferential object remains the target-level correlation with bootstrap confidence intervals.

Overall, the active-overlap evidence is not one of systematic alignment across all members. It is instead member-specific. Most members with sufficient local overlap are close to neutral, while a small number of members, especially member 96862, exhibit sizeable positive correlations with their own active client flow. We therefore interpret the analysis as a screening device for member-level episodes of prop–client alignment. Positive values are consistent with a front-running concern, but they are not proof of intent: common information, inventory management, client facilitation, and execution synchronisation can generate similar co-movement patterns.

6 Conclusion

Using trade-by-trade data for FTSE MIB constituents from 2024/06/03 to 2025/05/30, we reconstruct member-level metaorders and compare proprietary and client aggressive flow within a common framework. The main results are:

- **Basic properties differ mainly through aggressiveness rather than through radically different distributional shapes.** Both groups display broad, heavy-tailed distributions for duration, size, relative size, and participation rate, but proprietary metaorders are executed with systematically higher participation rates than client metaorders (Section 3, Table 4).

- **The metaorder sample is dominated by foreign executing members.** Foreign brokers account for almost the entire proprietary metaorder sample and for the large majority of client metaorders, so the aggregate results reported in the paper are primarily driven by foreign-member execution (Section 3.3, Table 5).

- **Price impact is concave for both groups, but the impact law differs sharply by trading capacity.** Client flow is close to the square-root benchmark, with $\hat{\gamma} = 0.5082 \pm 0.0228$, whereas proprietary flow is substantially more concave, with $\hat{\gamma} = 0.3662 \pm 0.0083$. In addition, the scale parameter is much larger for client flow ($Y_{\text{client}}/Y_{\text{prop}} \approx 1.76$), but this does not imply that the client fit lies uniformly above the proprietary one. Because the proprietary exponent is lower, the fitted proprietary curve lies above the client curve over most of the observed ϕ range and crosses it only around $\phi \approx 1.9 \times 10^{-2}$. (Section 4.2, Figure 6, Figure 7).

- **The dynamic response also differs across groups.** After execution, proprietary metaorders retain a larger fraction of peak impact than client metaorders: at $\tau = 3$, $\bar{I}(\tau{=}3)/\bar{I}(\tau{=}1) \approx 0.75$ for proprietary flow versus ≈ 0.57 for client flow. This suggests a more persistent component for proprietary impact and a larger transient component for client executions (Section 4.3, Figure 9).

- **Crowding is economically meaningful in both segments, but it is clearly stronger for client flow.** Within-group direction–imbalance correlations are positive and significant for both proprietary and client metaorders, equal to $r = 0.108$ (95% CI $[0.102, 0.113]$) and $r = 0.187$ (95% CI $[0.180, 0.196]$), respectively. Crowding also strengthens with participation rate, especially for clients, while cross-group relations remain weak and slightly contrarian. The all-vs-all correlations point in the same direction: client metaorders are more aligned with the aggregate market-side imbalance than proprietary metaorders (Section 5, Figure 11, Figure 12, Figure 13, Table 6). When daily crowding is linked directly to impact, more crowded metaorders do have larger impact, but the premium is larger for client flow; in pooled cell-level regressions, crowding enters positively while the proprietary dummy remains large and significant. A finer active-overlap analysis shows that proprietary metaorders are not exposed to a larger time-weighted count of overlapping metaorders, but their overlapping flow is more directionally aligned. Adding these active-overlap controls reduces the proprietary coefficient by about 16% in metaorder-level regressions. In the log-binned WLS specification, which mirrors the original power-law fit, the positive proprietary coefficient becomes smaller and statistically insignificant after active-overlap controls enter. Thus crowding explains a material part of the proprietary impact premium, especially through directional active overlap, but not through a simple count of overlapping metaorders (Section 5.5, Tables 7–10).

- **At the executing-member level, prop–client co-movement is small on average but not absent.** The mean per-member correlation is close to zero, yet the pooled correlation between proprietary direction and the imbalance of the same member's client flow is positive and statistically different from zero (0.011, 95% CI $[0.001, 0.022]$; permutation $p = 0.014$). The evidence is episodic and heterogeneous across members rather than uniform over time (Section 5.4, Figure 14, Figure 15). A more local active-overlap version of the same statistic confirms this heterogeneity: the pooled same-ISIN active correlation is small (0.036, 95% CI $[-0.032, 0.104]$), but one member exhibits a sizeable positive correlation with own active client flow ($r_m^{\text{act}} = 0.431$, 95% CI $[0.249, 0.614]$). Thus member-level prop–client alignment is not pervasive, but some members display recurrent local co-movement with their clients (Section 5.6, Figure 18).

Taken together, the results point to a coherent economic picture. Client metaorders are more crowded on the daily same-ISIN horizon and display a larger unconditional crowding premium, but the static impact ranking depends on the relative-size range considered: despite the larger client intercept Y, the more concave proprietary fit lies above the client fit over most of the observed ϕ range and crosses it only for sufficiently large ϕ. Client impact also decays more strongly after execution, whereas proprietary metaorders leave a more persistent footprint in prices. The active-overlap results suggest that part of this difference is associated with directional overlap during execution; the remaining gap is specification dependent, shrinking partially in metaorder-level regressions and disappearing in the lower-dimensional binned WLS check. The relevant heterogeneity is therefore not limited to size or execution speed alone: it also reflects differences in execution motives, information content, and interaction with the surrounding order-flow environment.

More broadly, the paper shows that proprietary and client metaorders should not be pooled into a single object. Pooling them would mask differences that matter for empirical impact measurement, for transaction-cost modelling, and for the interpretation of crowding and intermediation in the market. In this sense, the main contribution of the paper is not only to document distinct impact curves, but to show that who trades is itself an essential state variable for understanding how large orders move prices.

A Italian Executing Members

This appendix reports summary statistics for the subsample of member-level metaorders executed by Italian members (member nationality tag = IT). As discussed in Section 3.3, the proprietary IT subsample is small, so estimates should be interpreted cautiously.

Quantity	Proprietary (IT)	Client (IT)
Metaorders (N)	2093	24111
Buys / sells	1033 / 1060	13607 / 10504
Duration T (min): mean / median / IQR	52.74 / 42.44 / 58.84	39.44 / 25.16 / 45.35
Inter-arrival Δt (min): mean / median / IQR	4.87 / 0.39 / 3.77 (N=22676)	4.49 / 1.15 / 4.37 (N=211746)
Volume Q (shares): mean / median / IQR	49221 / 10950 / 25409	40465 / 8131 / 18792
Relative size Q/V: mean / median / IQR	0.0048 / 0.0014 / 0.0046	0.0030 / 0.0011 / 0.0022
Participation rate η: mean / median / IQR	0.0679 / 0.0185 / 0.0726	0.0635 / 0.0191 / 0.0650

Table 13: Summary statistics for member-level metaorders executed by Italian members (member nationality tag = IT). Values are mean / median / interquartile range (IQR). Inter-arrival times are computed across child-trade gaps (sample size N is the number of gaps).

References

[1] N. Torre. BARRA market impact model handbook. BARRA Inc., Berkeley, 1997.

[2] R. Almgren, C. Thum, E. Hauptmann, and H. Li. Direct estimation of equity market impact. Risk **18**(7), 58–62, 2005.

[3] R. F. Engle, R. Ferstenberg, and J. Russell. Measuring and modeling execution cost and risk. Chicago GSB Research Paper No. 08-09, 2008.

[4] I. Mastromatteo, B. Tóth, and J.-P. Bouchaud. Agent-based models for latent liquidity and concave price impact. Physical Review E **89**(4), 042805, 2014.

[5] X. Brokmann, E. Serie, J. Kockelkoren, and J.-P. Bouchaud. Slow decay of impact in equity markets. arXiv preprint arXiv:1407.3390, 2014.

[6] E. Bacry, A. Iuga, M. Lasnier, and C.-A. Lehalle. Market impacts and the life cycle of investors orders. arXiv preprint arXiv:1412.0217, 2014.

[7] O. E. Barndorff-Nielsen, P. R. Hansen, A. Lunde, and N. Shephard, Designing realized kernels to measure the ex post variation of equity prices in the presence of noise, Econometrica, 76(6):1481–1536, 2008.

[8] Tsaknaki, I. Y., Lillo, F., and Mazzarisi, P. (2025). Online learning of order flow and market impact with Bayesian change-point detection methods. *Quantitative Finance*, 25(2), 307–322.

[9] A. Clauset, C. R. Shalizi, and M. E. J. Newman, Power-law distributions in empirical data, SIAM Review, 51 (2009), pp. 661–703, `https://doi.org/10.1137/070710111`.

[10] J.-P. Bouchaud, J. D. Farmer, and F. Lillo. How markets slowly digest changes in supply and demand. In T. Hens and K. R. Schenk-Hoppé (eds.), Handbook of Financial Markets: Dynamics and Evolution. Academic Press, 2008.

[11] E. Moro, J. Vicente, L. G. Moyano, A. Gerig, J. D. Farmer, G. Vaglica, F. Lillo, and R. N. Mantegna. Market impact and trading profile of hidden orders in stock markets. Physical Review E **80**(6), 066102, 2009.

[12] G. Vaglica, F. Lillo, and R. N. Mantegna. Statistical identification with hidden Markov models of large order splitting strategies in an equity market. New Journal of Physics **12**(7), 075031, 2010.

[13] C. Gomes and H. Waelbroeck. Is market impact a measure of the information value of trades? Market response to liquidity vs. informed metaorders. Quantitative Finance **15**(5), 773–793, 2015.

[14] E. Zarinelli, M. Treccani, J. D. Farmer, and F. Lillo. Beyond the square root: Evidence for logarithmic dependence of market impact on size and participation rate. Market Microstructure and Liquidity, **1**(2):1550004, 2015.

[15] Sias, R. W. (2004). Institutional herding. Review of Financial Studies, 17(1), 165–206. doi:10.1093/rfs/hhg035

[16] Dasgupta, A., Prat, A., and Verardo, M. (2011). The price impact of institutional herding. Review of Financial Studies, 24(3), 892–925. doi:10.1093/rfs/hhq137

[17] Lou, D. (2012). A flow-based explanation for return predictability. Review of Financial Studies, 25(12), 3457–3489. doi:10.1093/rfs/hhs103

[18] F. Bucci, M. Benzaquen, F. Lillo, and J.-P. Bouchaud. Slow decay of impact in equity markets: insights from the ANcerno database. Market Microstructure and Liquidity **4**(3–4), 1950006, 2018.

[19] M. Briere, C.-A. Lehalle, T. Nefedova, and A. Raboun. Modelling transaction costs when trades may be crowded: a Bayesian network using partially observable orders imbalance. Book chapter, 2019.

[20] Puckett, A. and Yan, X. (2008). Short-term institutional herding and its impact on stock prices. *SSRN Electronic Journal*. DOI: 10.2139/ssrn.972254

[21] F. Bucci, I. Mastromatteo, Z. Eisler, F. Lillo, J.-P. Bouchaud, and C.-A. Lehalle. Co-impact: crowding effects in institutional trading activity. Quantitative Finance **20**(2), 193–205, 2020.

[22] F. Bucci, F. Lillo, J.-P. Bouchaud, and M. Benzaquen. Are trading invariants really invariant? Trading costs matter. Quantitative Finance **20**(7), 1059–1068, 2020.

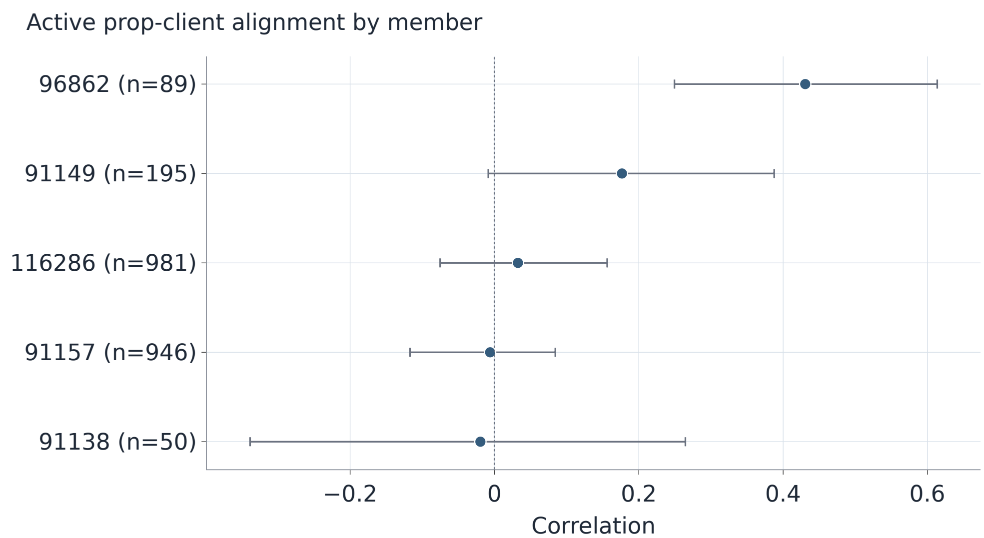
[23] Y. Sato and K. Kanazawa. Does the square-root price impact law belong to the strict universal scalings? Quantitative support by a complete survey of the Tokyo Stock Exchange market. Preprint, 2024.

[24] G. Maitrier, G. Loeper, and J.-P. Bouchaud. Generating realistic metaorders from public data. arXiv preprint arXiv:2503.18199, 2025.

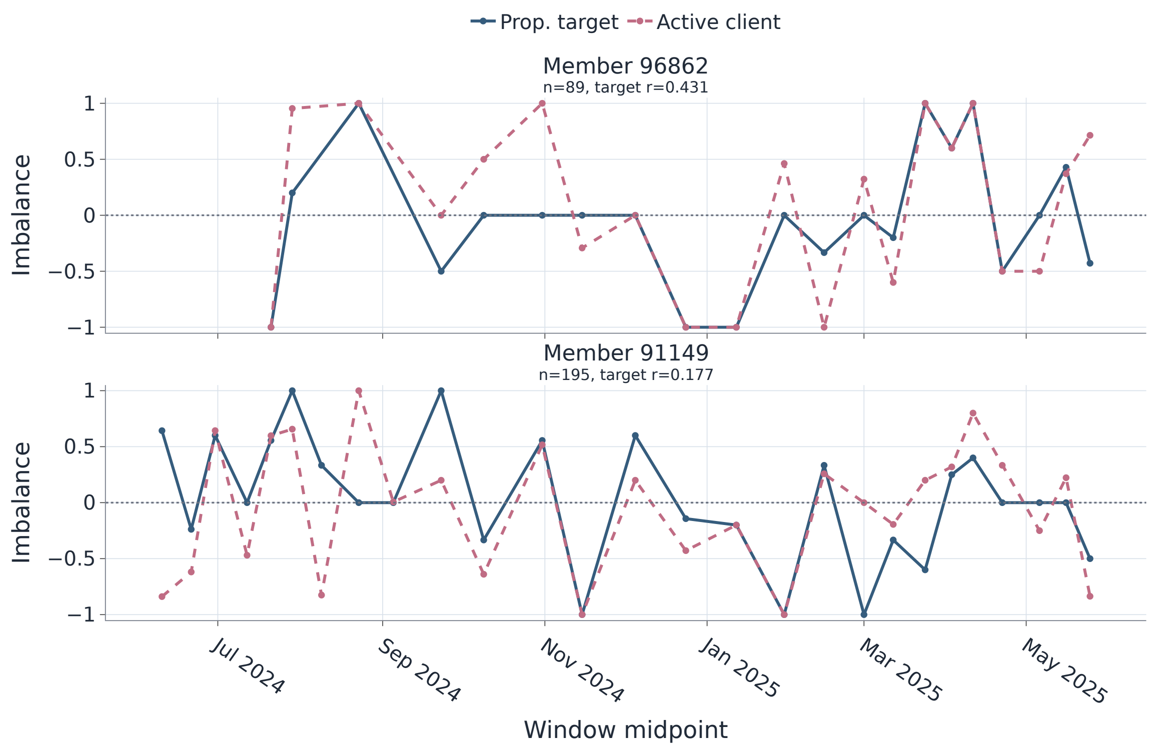
[25] M. Naviglio, G. Bormetti, F. Campigli, G. Rodikov, and F. Lillo. Why is the estimation of metaorder impact with public market data so challenging? arXiv preprint arXiv:2501.17096, 2025.

[26] G. Guillaume. The hidden complexity of price formation. PhD thesis, 2025.

[27] D. M. Di Nosse. Metaorders_PriceImpact: Study of metaorder price impact using proprietary CONSOB data. GitHub repository, 2025. Available at `https://github.com/DanieleMDiNosse/Metaorders_PriceImpact`.



(a) Per-member correlations, same-ISIN all-active environment. Points show r_m^{act} and horizontal bars show 95% bootstrap confidence intervals.



(b) Window-level co-movement for the two strongest positive members.

Figure 18: Active member-level prop–client alignment. The top panel reports r_m^{act} for members with at least 30 valid proprietary targets. The bottom panel compares the proprietary target-direction imbalance with the active client imbalance for members 96862 and 91149 in non-overlapping 5-trading-day windows.